

BIOMECHANICS (EPHE 341)

Forces

Optional reading: McGinnis Chpt 1 pg 20–
32, 39

Force

Biomechanics is the study of the effects of internal and external **forces** on the human body in movement and rest.

Simply defined a force is a **push or a pull**.

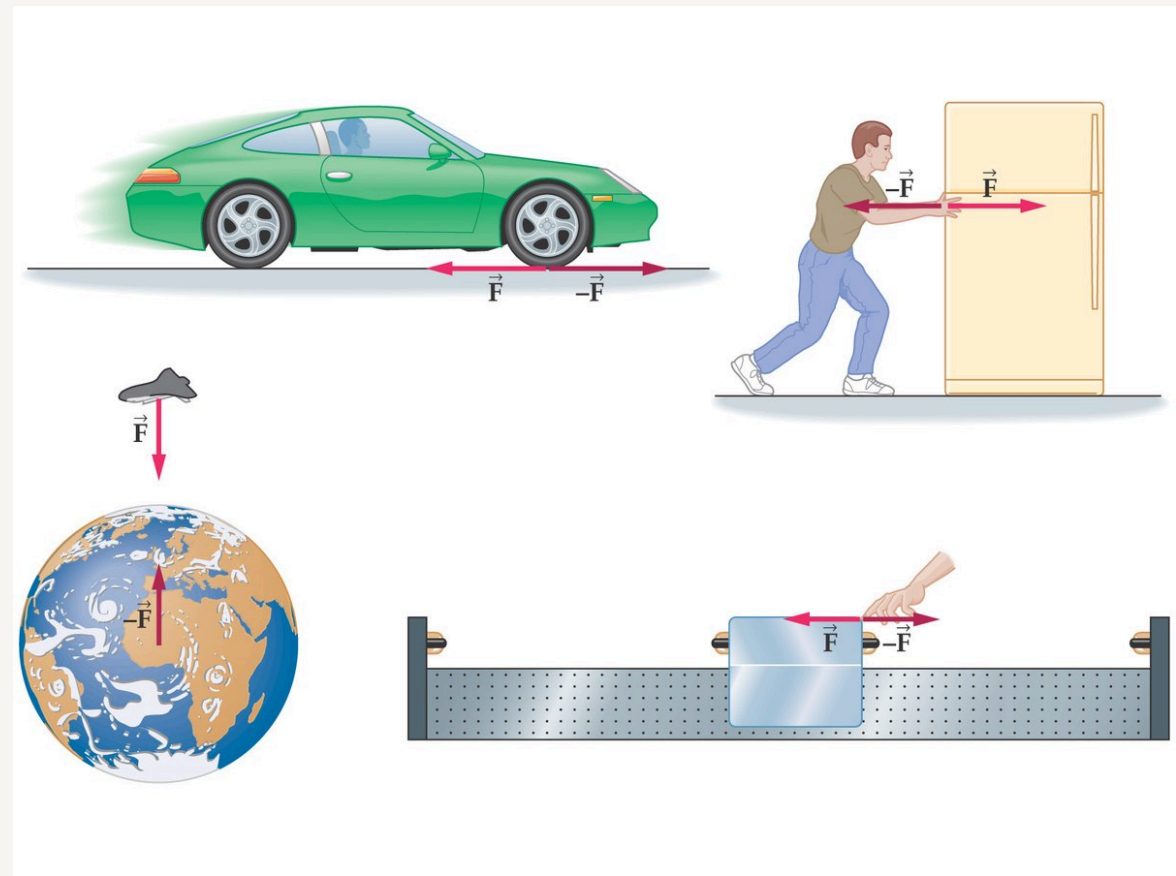
Force is a **vector** quantity

Can cause a change in the state of motion of an object.

start, stop, speed up, slow down or change direction

Can either accelerate and/or deform an object

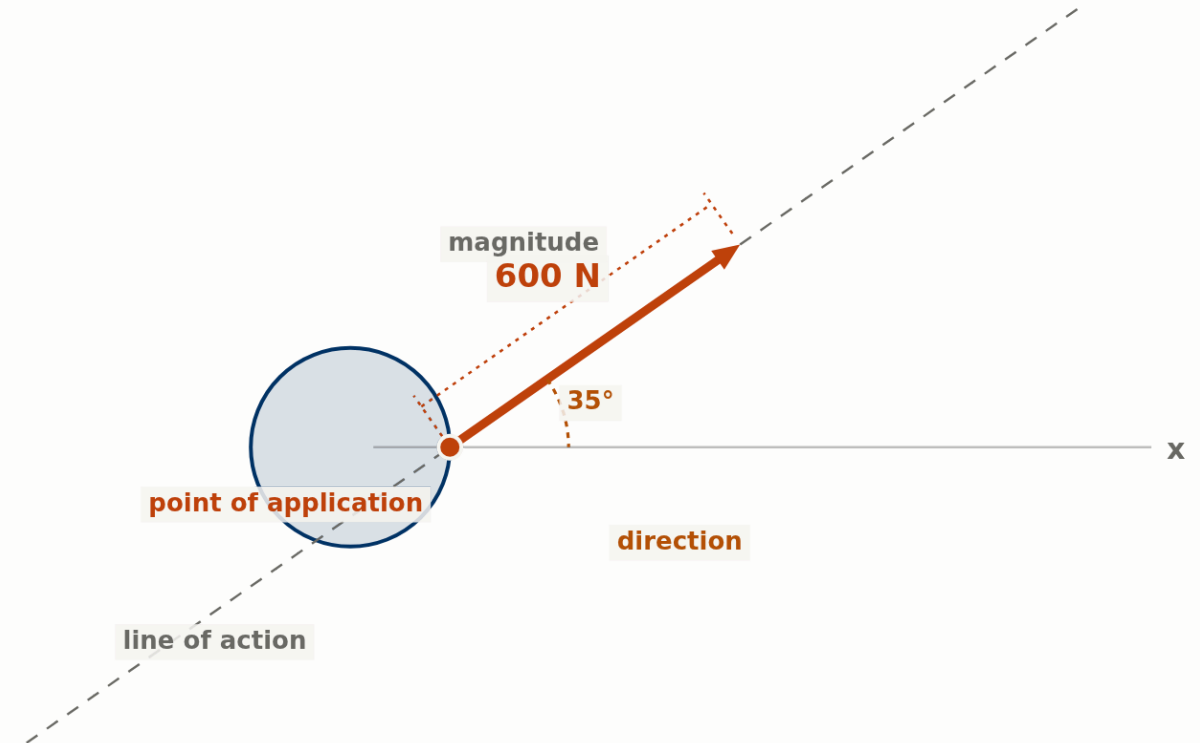
comes in pairs: action and reaction



Force vectors

Some important points about force vectors

- **Magnitude** – length of the arrow
- **Direction** – relative to an axis
- **Line of action** – the shaft of the arrow
- **Point of application** – where it acts on an object



600 N at **35°** above the x axis · applied at the **centre**

Magnitude changes the length, direction swings the whole line of action, and the point of application decides where on the object it acts. Move it off the centre.

through the centre

at the top edge

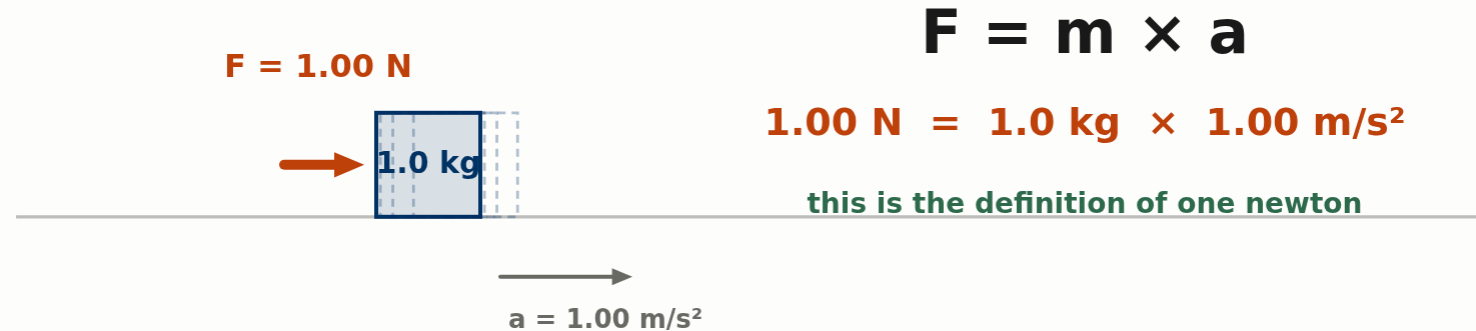
at the side

Magnitude  **600 N**

Force — units

The Newton (N) is the SI unit of measurement for force. One N of force is defined as: the force required to accelerate a 1 kg mass at 1 m/s². $1 \text{ N} = (1.0 \text{ kg})(1.0 \text{ m/s}^2)$

The equation for force (F) is $\mathbf{F = ma}$, where **m** is mass and **a** is acceleration.



$$1.00 \text{ N} = 1.0 \text{ kg} \times 1.00 \text{ m/s}^2$$

One newton is the force that accelerates one kilogram at one metre per second squared — set both sliders to 1 and the equation says so. Doubling either one doubles the force.

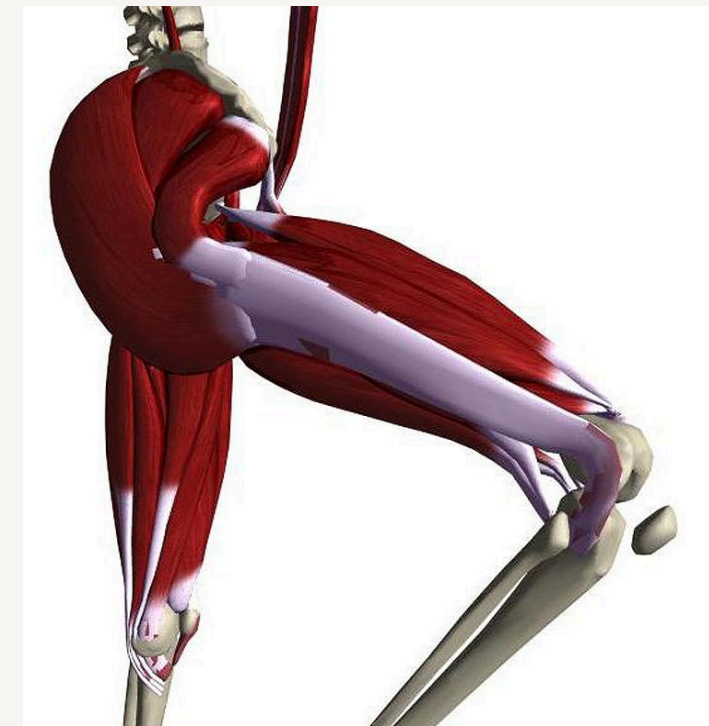
Mass	<input type="range"/>	1.0 kg
Acceleration	<input type="range"/>	1.00 m/s ²

Internal forces

Internal forces act within an object or system.

Compressive (pushing) and **tensile** (pulling) forces.

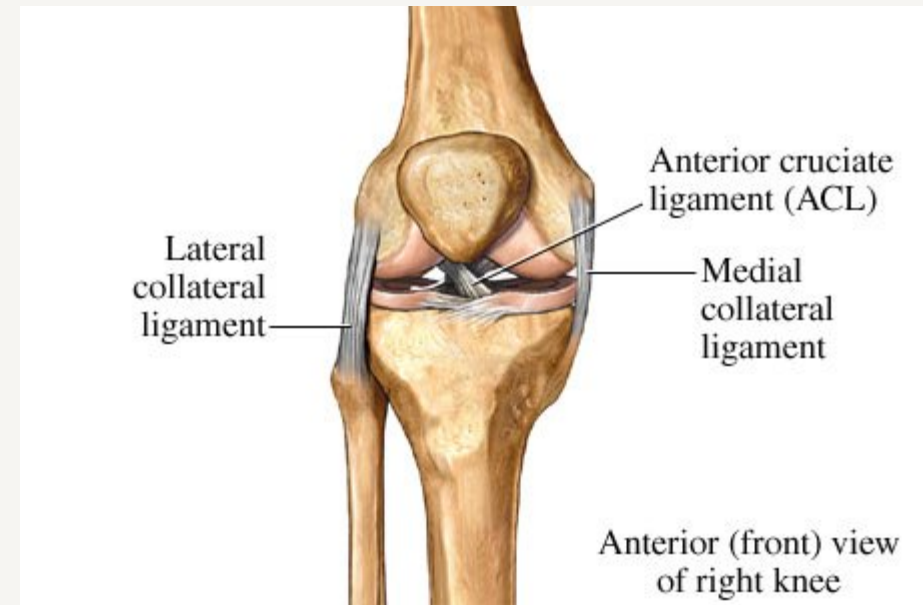
When you land from a jump there is a force from your lower leg on your upper leg through your knee and an equal and opposite force from your upper leg to your lower leg through your knee.



Measuring internal forces in biomechanics

It is difficult to directly measure internal forces in human biomechanics.

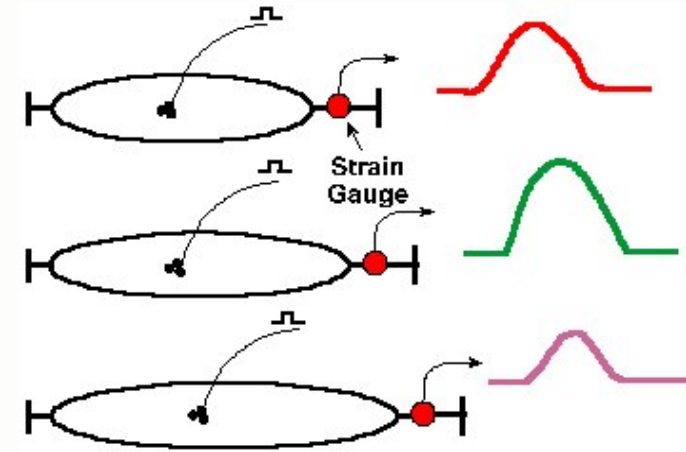
An invasive option is to use **strain gauges in cadaver limbs** to conduct testing of injury occurrence and prevention.



Measuring internal forces in biomechanics

Strain gauge or optical fiber on/in human or animal tendon can be used to measure the **magnitude** of muscle forces.

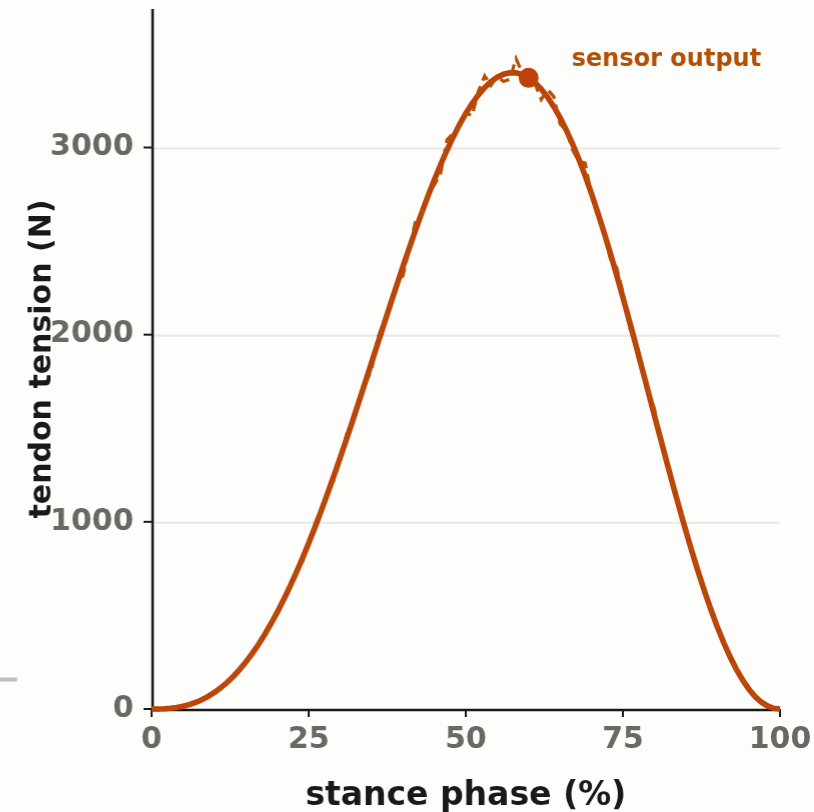
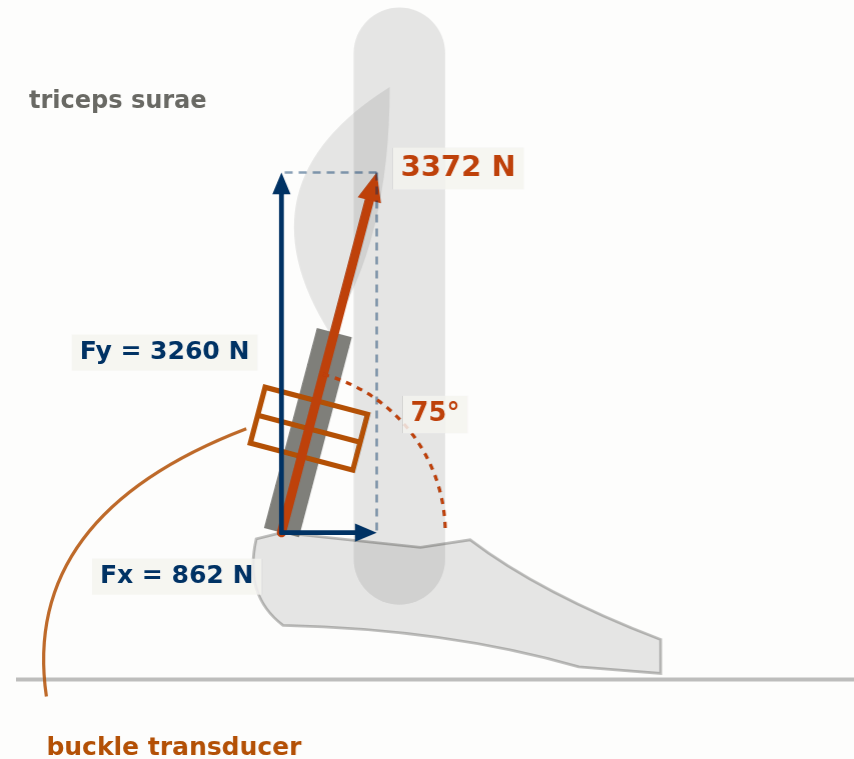
But what about the **direction**?



Measuring internal forces in biomechanics

A buckle transducer or an optical fibre in the Achilles tendon

The sensor gives the **magnitude** of the tension in the tendon. Change the ankle angle and that number does not move — but the line the tendon pulls along does, and with it every component of the force at the joint. The other half of the vector has to come from the anatomy.



Internal force vector resolution

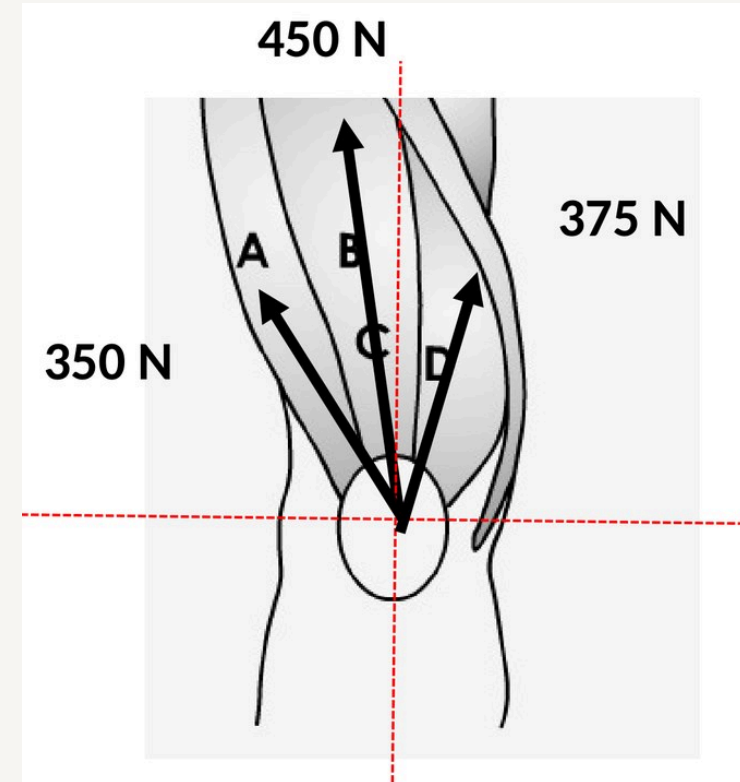
Step 1: Determine what the question is asking for · Step 2: Draw a diagram with all known values

Using fibre optic force measurement we have determined the muscle forces of the quadriceps during a squat. Find the resultant quadriceps muscle force given the individual muscle forces of vastus lateralis (VL), vastus intermedialis (VI) and vastus medialis (VM)

$$VL = 350 \text{ N} \quad 60^\circ$$

$$VI = 450 \text{ N} \quad 80^\circ$$

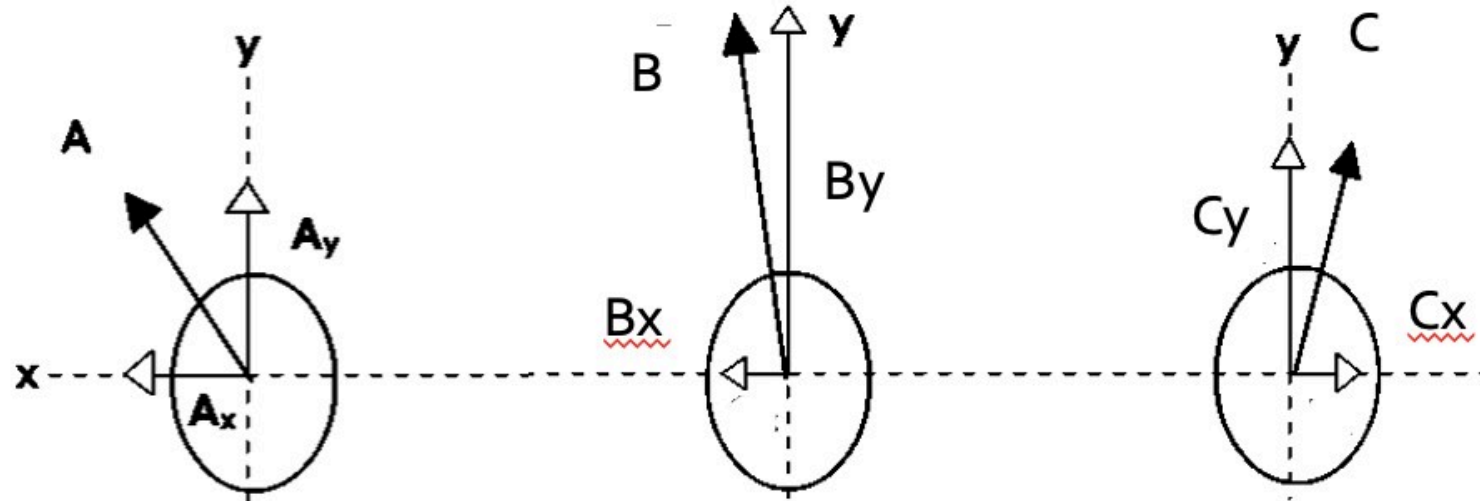
$$VM = 375 \text{ N} \quad 80^\circ$$



Internal force vector resolution

Step 3: Isolate the forces and determine force components using trig functions

It is easiest to isolate each muscle force and then determine the separate components.



Step 3: Isolate the forces and determine force components using trig functions

$$VL_x = -350\cos 60^\circ = \mathbf{-175\text{ N}}$$

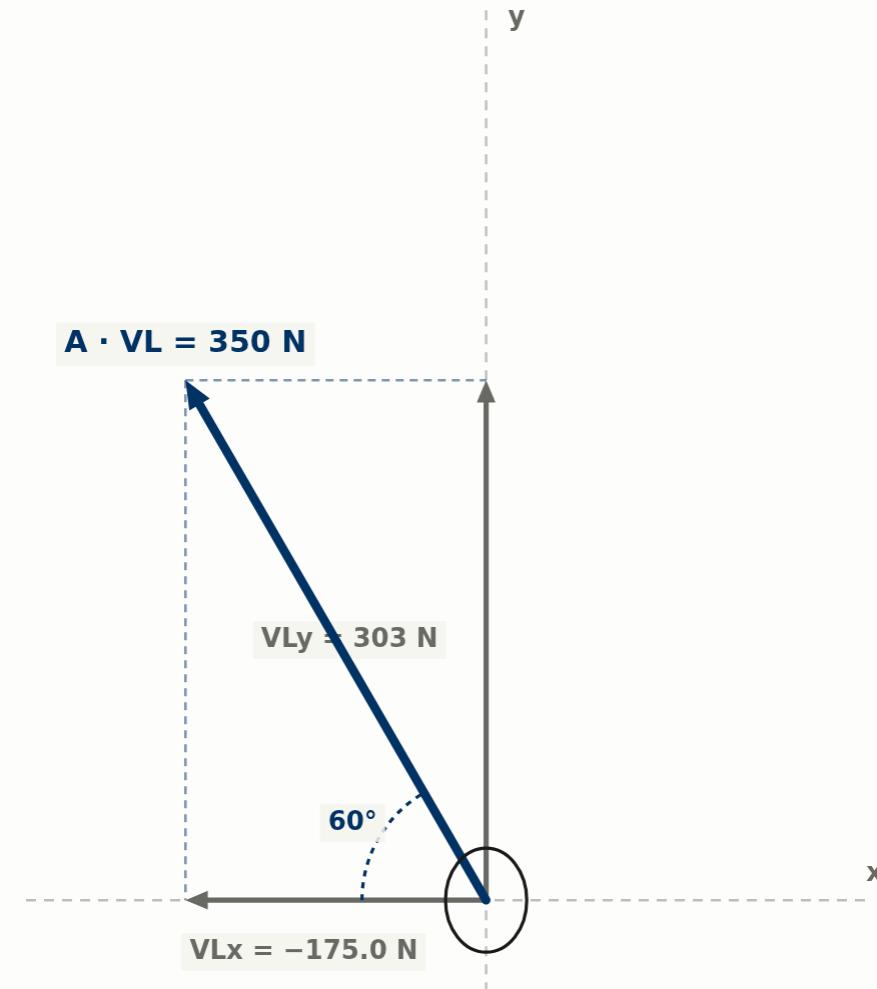
$$VL_y = 350\sin 60^\circ = \mathbf{303\text{ N}}$$

$$VI_x = -450\cos 80^\circ = \mathbf{-78.1\text{ N}}$$

$$VI_y = 450\sin 80^\circ = \mathbf{443\text{ N}}$$

$$VM_x = 375\cos 80^\circ = \mathbf{65.1\text{ N}}$$

$$VM_y = 375\sin 80^\circ = \mathbf{369\text{ N}}$$



$$VL = 350\text{ N at } 60^\circ \cdot F_x = \mathbf{-175.0\text{ N}}, F_y = \mathbf{303\text{ N}}$$

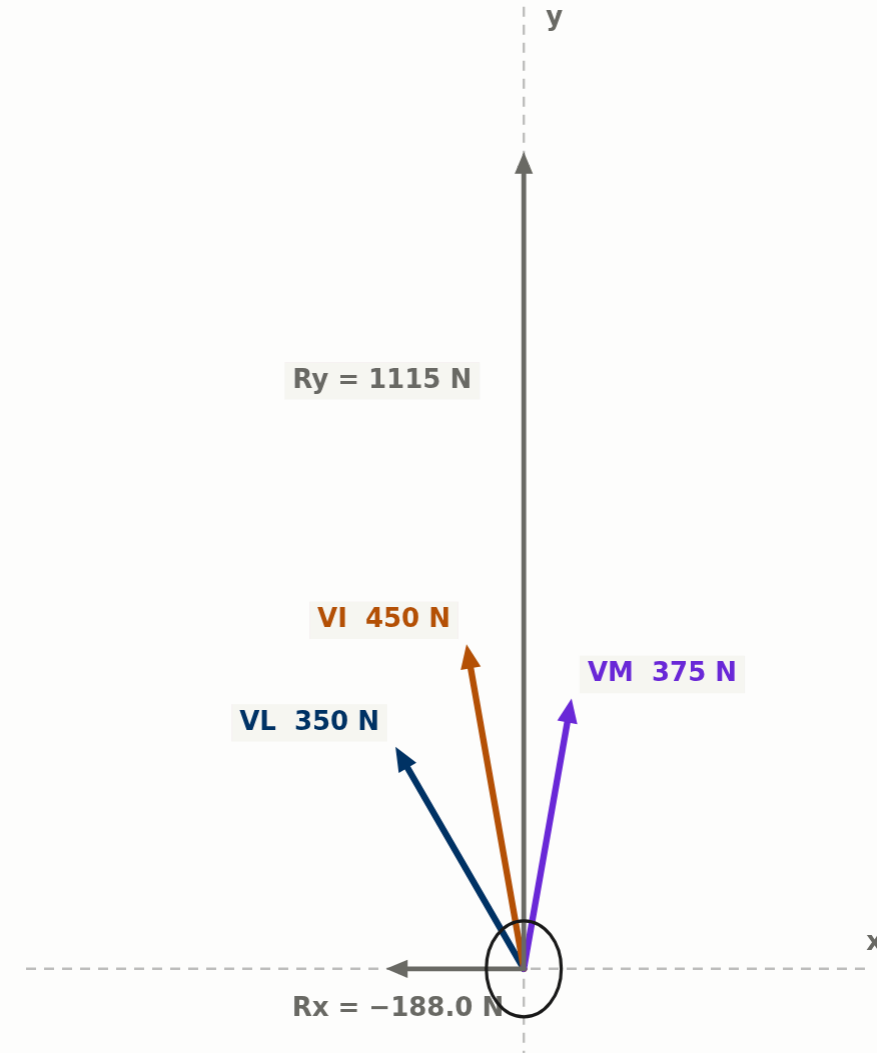
Internal force vector resolution

Step 4: Add the respective force components to determine the resultant vector component

You could simply add the forces or use a table similar to when adding linear kinematic variables

$$\begin{aligned} R_x &= \Sigma \text{ Horizontal Forces} \\ &= -175 \text{ N} - 78.1 \text{ N} + 65.1 \text{ N} = \\ &\quad \mathbf{-188.8 \text{ N}} \end{aligned}$$

$$\begin{aligned} R_y &= \Sigma \text{ Vertical Forces} \\ &= 303 \text{ N} + 443 \text{ N} + 369 \text{ N} = \mathbf{1115} \\ &\quad \mathbf{N} \end{aligned}$$

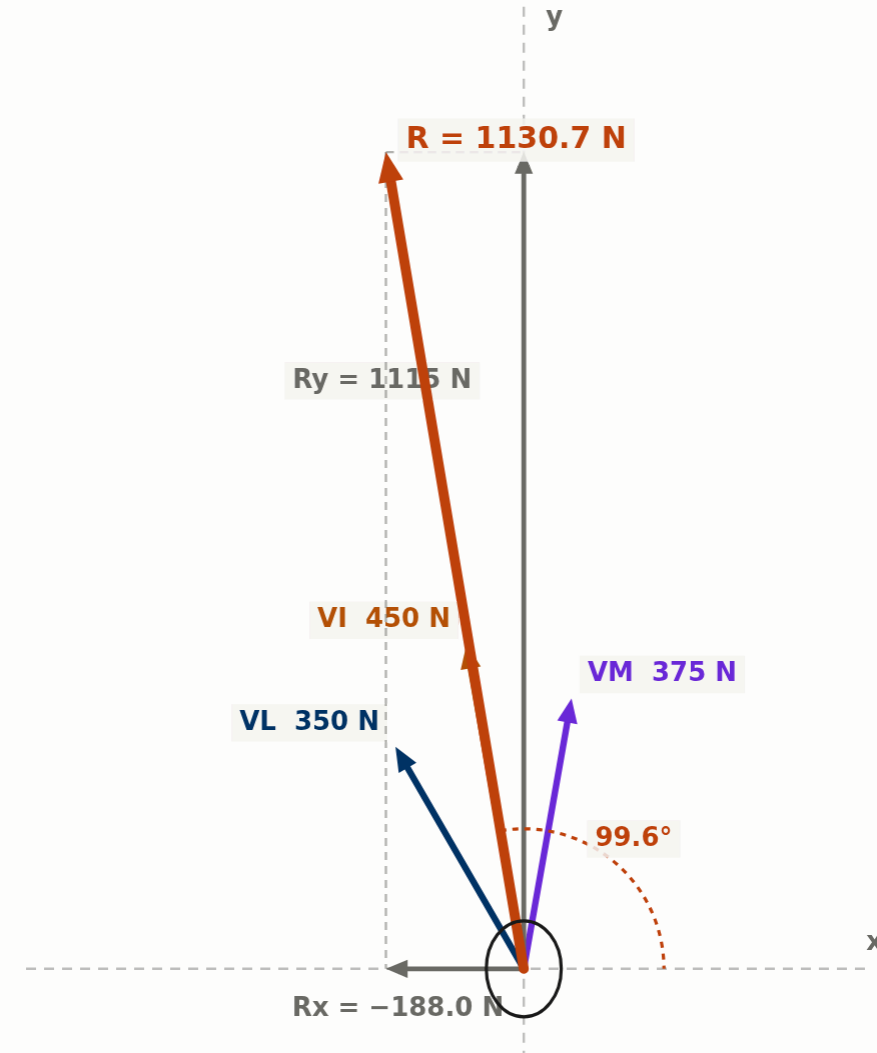


$$R = \mathbf{1130.74 \text{ N}} \text{ at } \mathbf{99.6^\circ} \cdot R_x = -188.0 \text{ N}, R_y = 1115 \text{ N}$$

Step 5: Solve for the resultant force magnitude and direction

$$\begin{aligned} R^2 &= R_x^2 + R_y^2 \\ &= -188.8^2 + 1115^2 \\ R &= \mathbf{1130.87 \text{ N}} \end{aligned}$$

$$\begin{aligned} \tan \theta &= R_y / R_x \\ \theta &= \tan^{-1} (R_y / R_x) \\ \theta &= \tan^{-1} (1115 / -188.8) \\ &= \mathbf{-80.4} \end{aligned}$$



$R = \mathbf{1130.74 \text{ N}}$ at $\mathbf{99.6^\circ}$ · $R_x = -188.0 \text{ N}$, $R_y = 1115 \text{ N}$

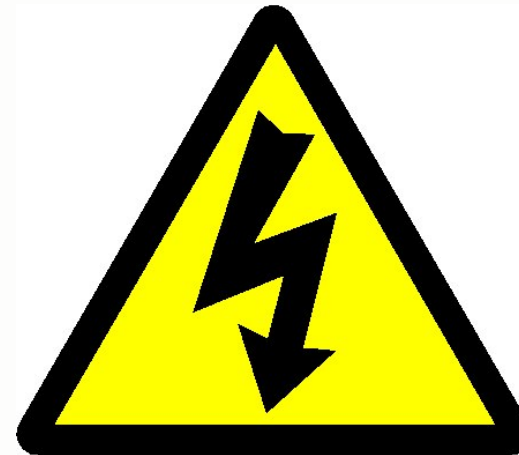
External forces

Forces that act on an object as a result of the surroundings.

Contact and non-contact.

Non-contact forces

- Gravity
- Magnetic
- electrical



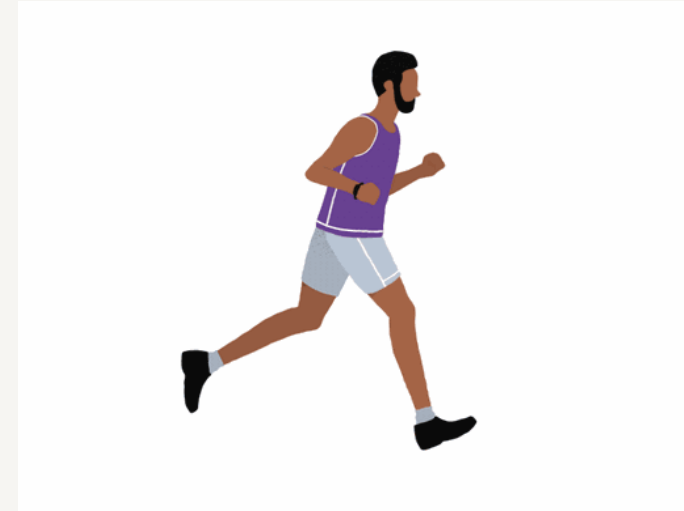
Gravity and contact forces

Contact forces

Occur between objects in contact with each other — solid or fluid. Air resistance and water resistance are fluid contact forces.

Occur in pairs: **action and reaction**.

The force of gravity acting on us causes us to contact the ground.

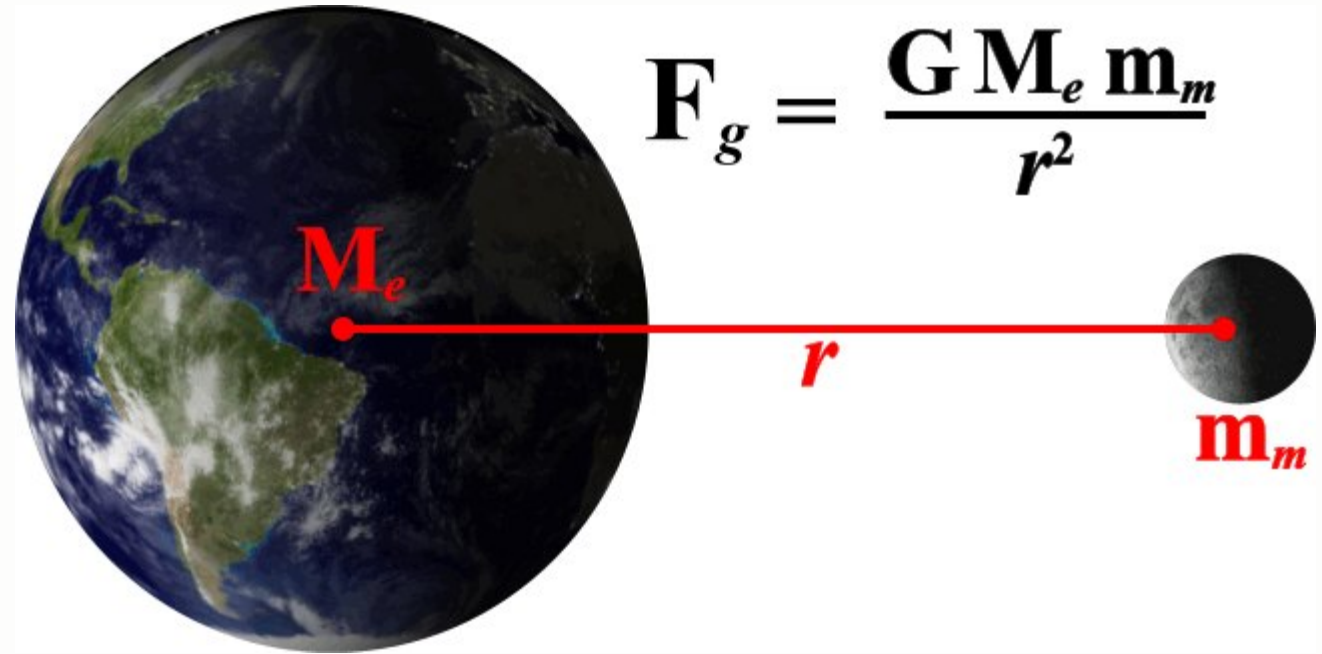


Gravity

We experience gravity as **weight**.

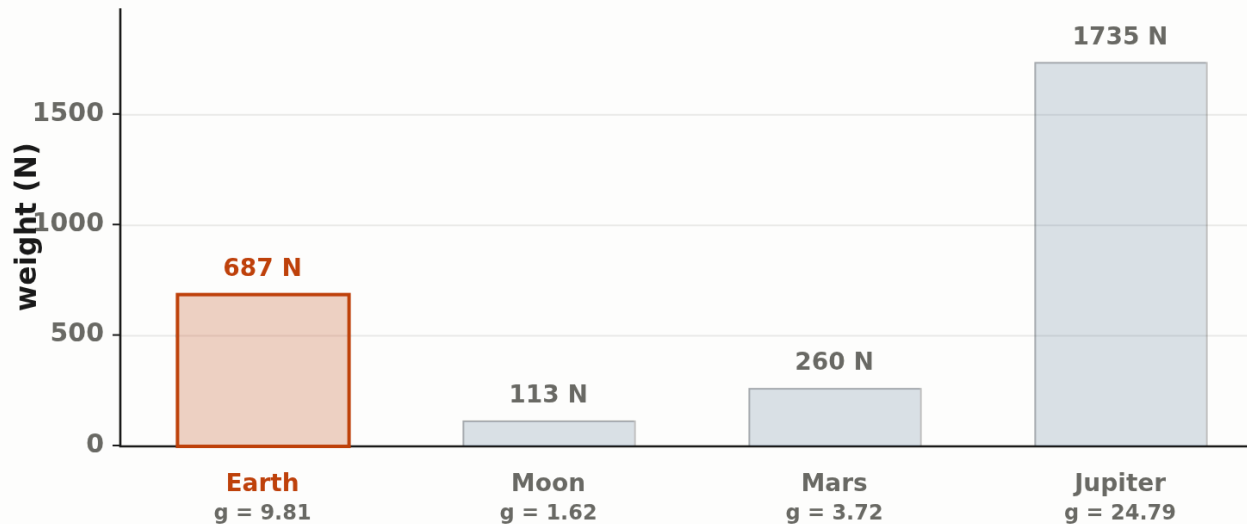
All objects with mass have an attraction to other objects with mass. The magnitude of the attraction depends on the mass of each object and the distance between the two objects. The earth attracts someone and the person attracts the earth with an equal but opposite force.

We have a constant force acting on us pointed towards the centre of the earth acting through our **centre of mass**.



Gravity and weight

The weight of an object is defined as the force of gravity acting on an object. We know that the acceleration due to gravity is 9.81 m/s^2 , and the equation for force is $F = ma$.



$$W = mg = 70 \times 9.81 = 687 \text{ N} \cdot \text{ on Earth}$$

The mass never changes — 70 kg is 70 kg anywhere. The weight is a **force**, and it changes with whatever is pulling on you. On Earth, multiply kilograms by about ten to get newtons.

Body mass 70 kg

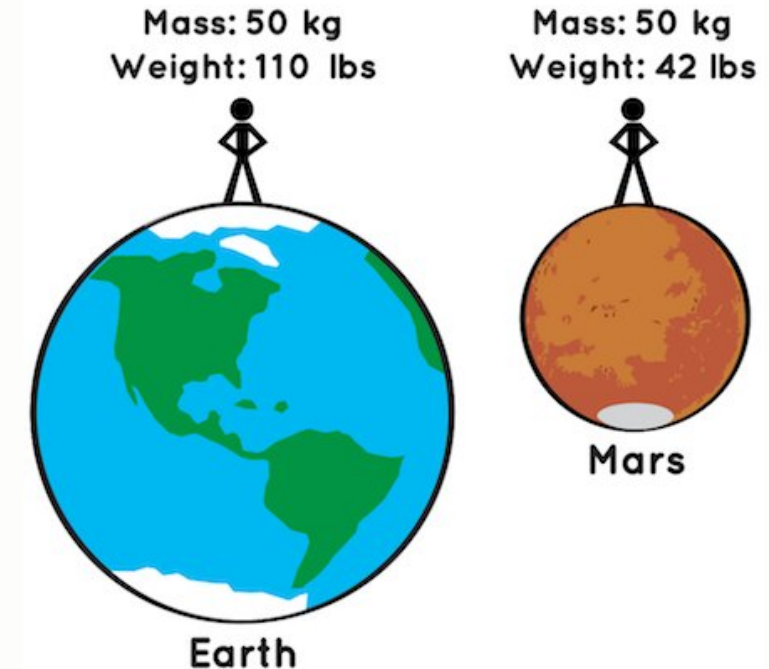
Earth

Moon

Mars

Jupiter

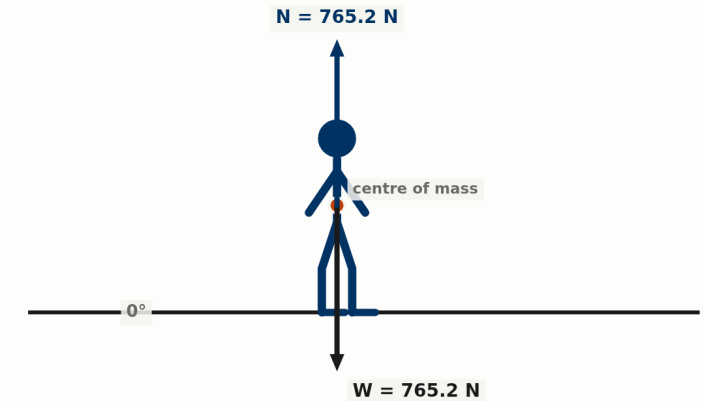
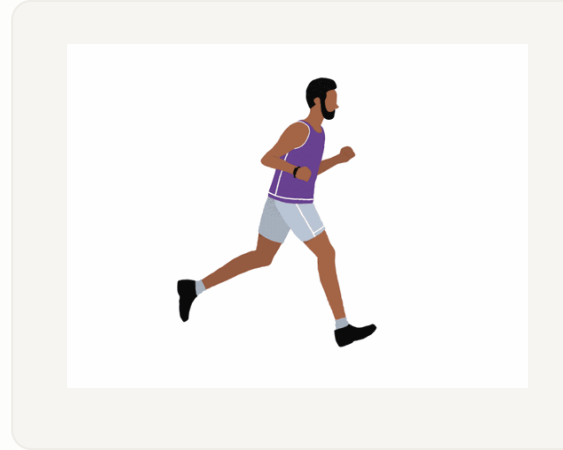
Therefore your weight (W) is equal to your body mass in kilograms (m) multiplied by the acceleration due to gravity ($g = 9.81 \text{ m/s}^2$), $W = mg$. **What is your weight in Newtons?**



Force of gravity and normal force

The force of gravity is acting towards the earth through our center.

This causes us to contact the ground and the ground must exert a force to hold us up just as we must exert muscular force to remain upright.



The component of the force of the ground acting on us perpendicular to the contact surface is called the **normal force**.

When we are on a surface parallel to the horizontal, the normal force is equal in magnitude to the force of gravity and opposite in direction....

Flat ground: the normal force is **equal and opposite** to the weight, 765.2 N.

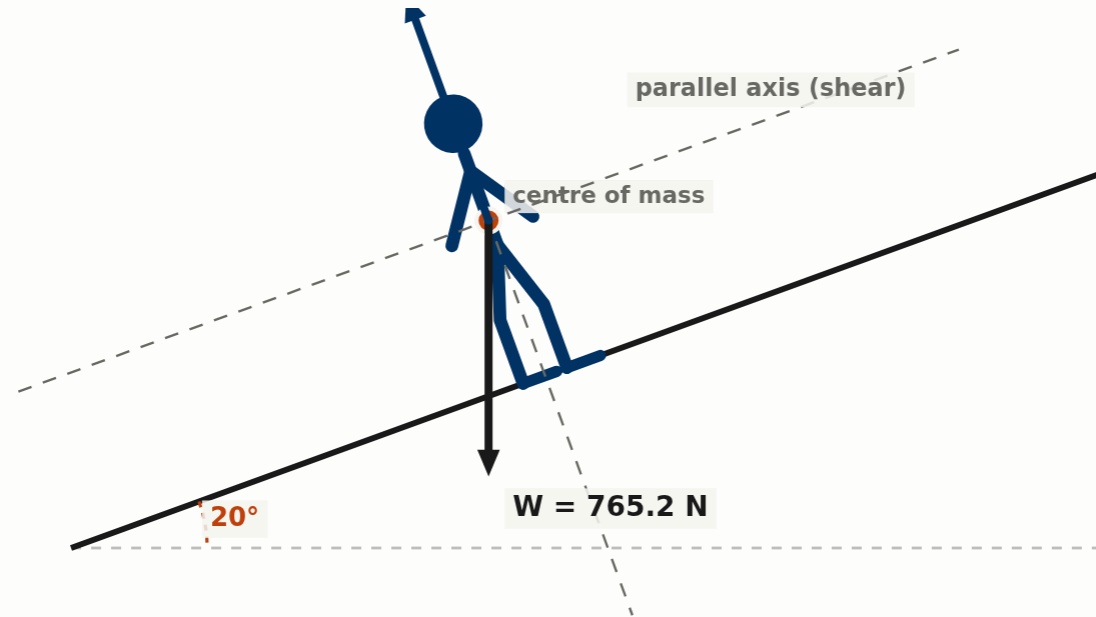
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Force of gravity and normal force

When we or an object are on a **slanted surface**:

The force of gravity vector still points directly towards the centre of the earth but the normal force stays perpendicular to the contact surface and is no longer equal and opposite to the magnitude and direction of the force of gravity.

So how do we calculate the normal force?



At 20° the normal force is **719.0 N**, not 765.2 N — the missing 46.1 N has become **261.7 N of shear** down the slope

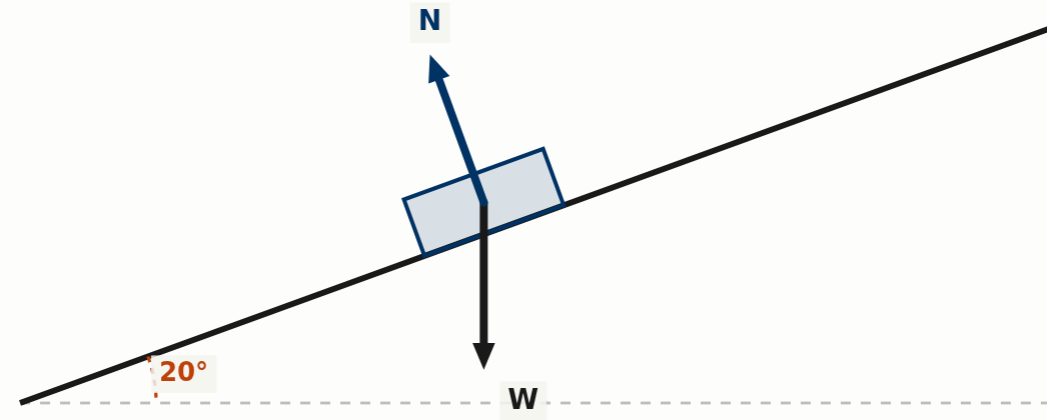
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Angle of the
surface



20°

With contact forces it is customary to create an axis system that is **parallel and perpendicular to the contact surface**.



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

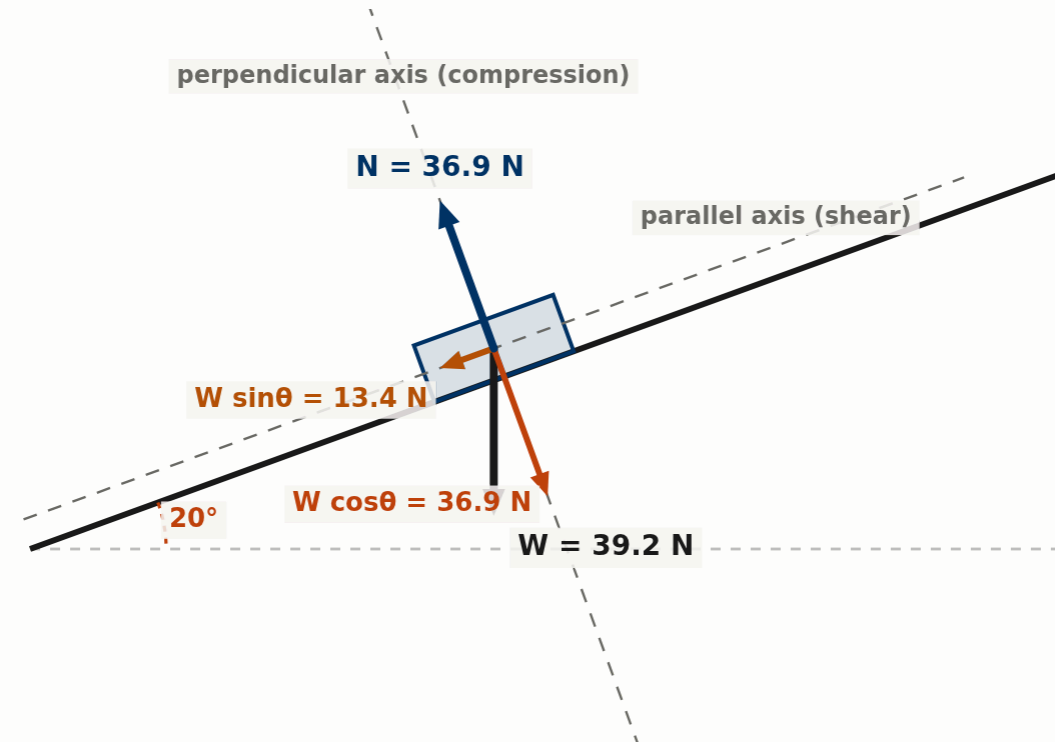
Add the axes

Show the components

Then any forces acting in the system can be resolved into components that are:

- Perpendicular to the contact surface (**compression forces**) and
- Parallel to the contact surface (**shear forces**).

The normal force is equal and opposite to the component of gravity force that is perpendicular to the contact surface.



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

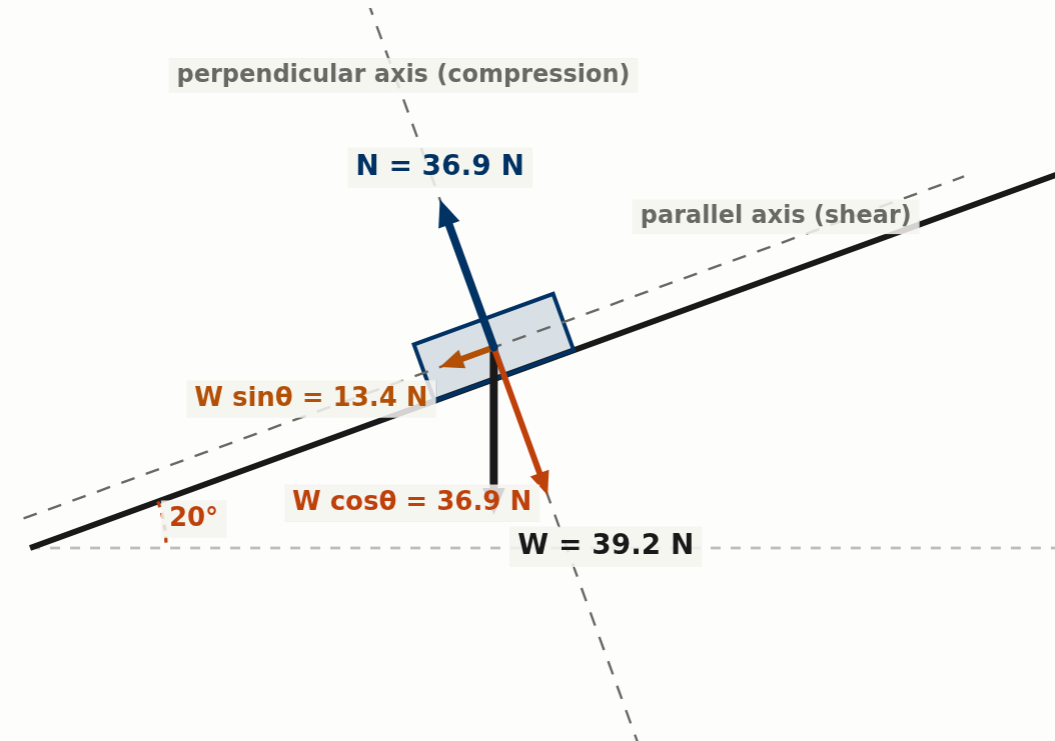
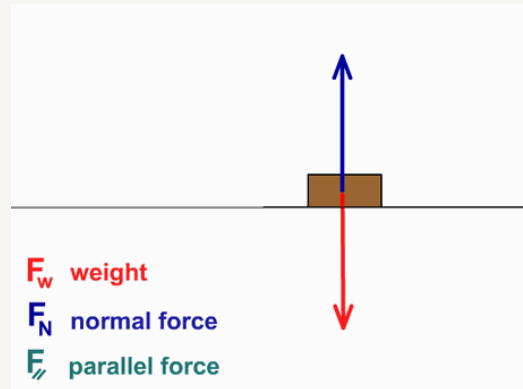
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

► Sweep the surface angle

What would happen to:

- The force of gravity
- Normal force
- and force components

If we change the angle of the contact surface?

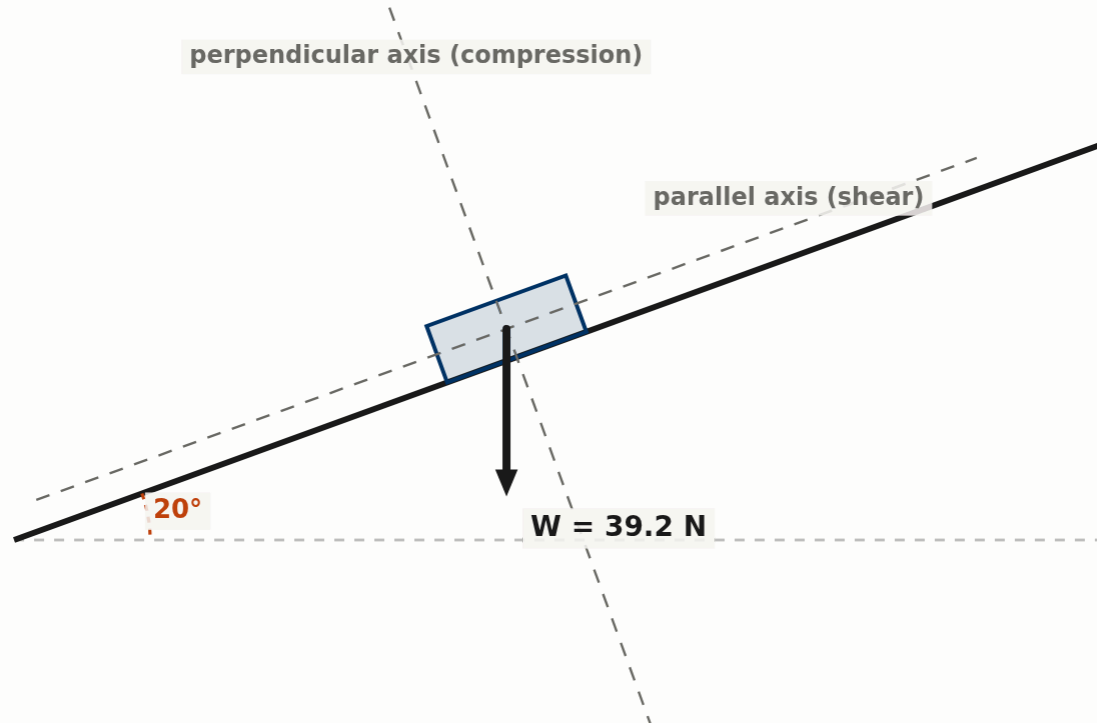


At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

► Sweep the surface angle

How to resolve for the components of force on a slant surface and solve for the normal force. We need to know the angle of the slant (20°) and the mass of the object (4 kg).



4.0 kg on a 20° slope

the weight

$$W = mg = 4.0 \times 9.81 = \mathbf{39.24 \text{ N}}$$

N
36.9 N

shear
13.4 N

N / W
94%

At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

4th — solve for the parallel and perpendicular components of force.

Parallel component

$$\text{Opposite} = H \cdot \sin 20^\circ$$

$$\text{Opposite} = 39.24 \cdot \sin 20^\circ$$

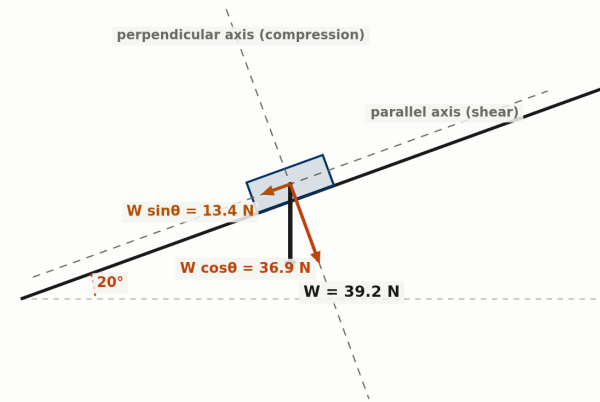
$$\text{Opposite} = 13.42 \text{ N}$$

Perpendicular component

$$\text{adjacent} = H \cdot \cos 20^\circ$$

$$\text{adjacent} = 39.24 \cdot \cos 20^\circ$$

$$\text{adjacent} = 36.87 \text{ N}$$



4.0 kg on a 20° slope

the weight

$$W = mg = 4.0 \times 9.81 = \mathbf{39.24 \text{ N}}$$

parallel — sliding down the surface

$$W \sin \theta = 39.24 \cdot \sin 20^\circ = \mathbf{13.42 \text{ N}}$$

perpendicular — pressed into the surface

$$W \cos \theta = 39.24 \cdot \cos 20^\circ = \mathbf{36.87 \text{ N}}$$

N
36.9 N

shear
13.4 N

N / W
94%

At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Force

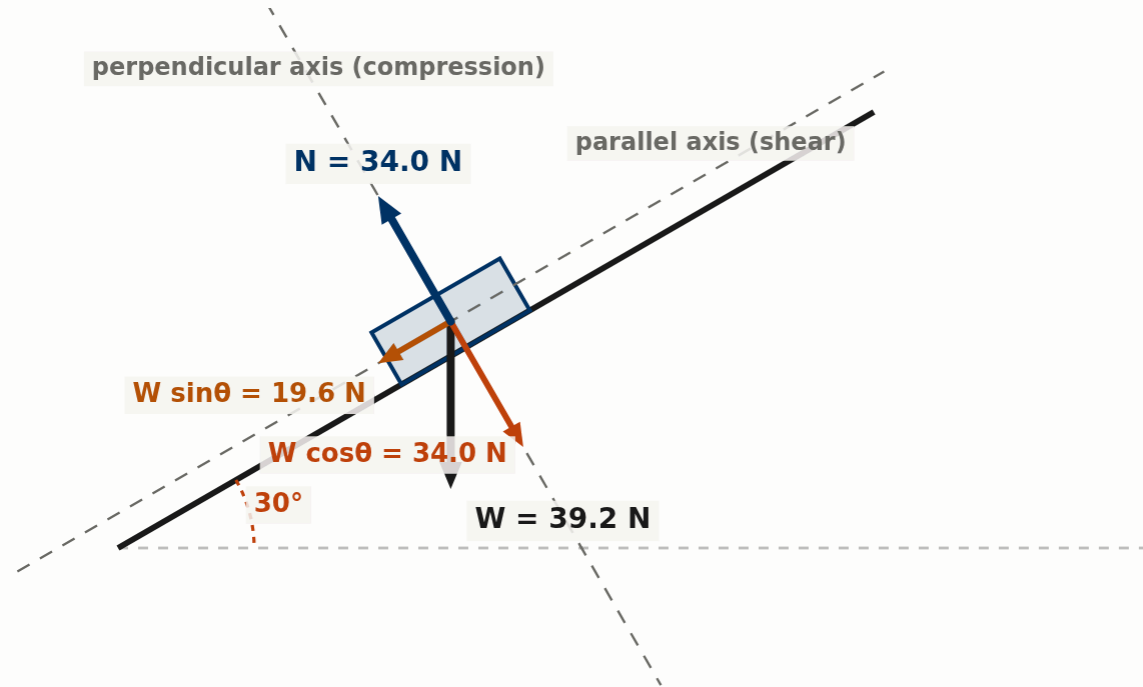
In general there is a simple rule to determine the parallel and perpendicular components of force.

For any incline of a certain angle the parallel and perpendicular components of force will be:

$$\text{parallel} = mg \sin \theta$$

$$\text{perpendicular} = mg \cos \theta$$

Where mg is the equation for the weight of the object.



At 30° the normal force is **34.0 N**, not 39.2 N — the missing 5.3 N has become **19.6 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Angle of the
surface



30°

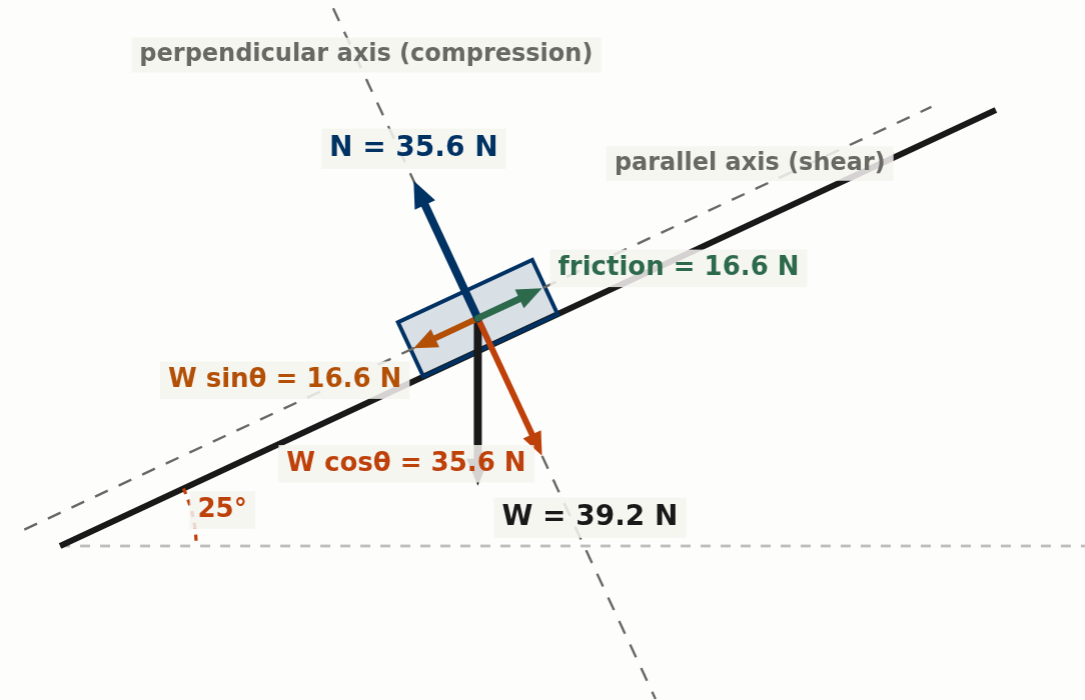
Friction

The parallel component of gravity will tend to pull the block down the surface...

Friction force is required to be equal and opposite of the parallel force or the block would slide down the slope.

Friction force is proportional to the normal force and acts perpendicular to it.

We will discuss more about friction in a following lecture.



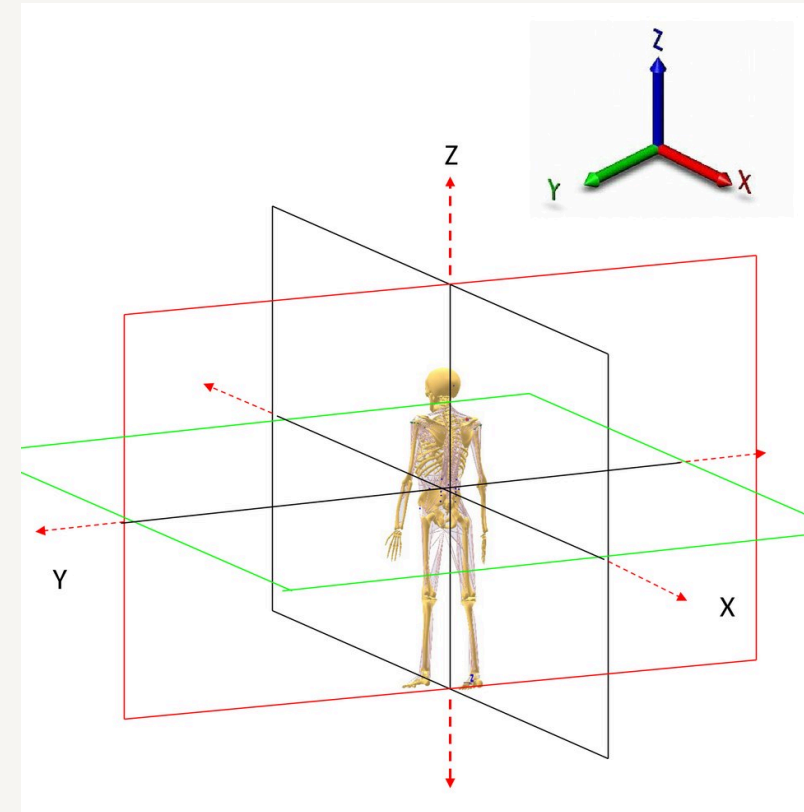
At 25° the normal force is **35.6 N**, not 39.2 N — the missing 3.7 N has become **16.6 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens, and why friction has to hold up whatever the shear is.

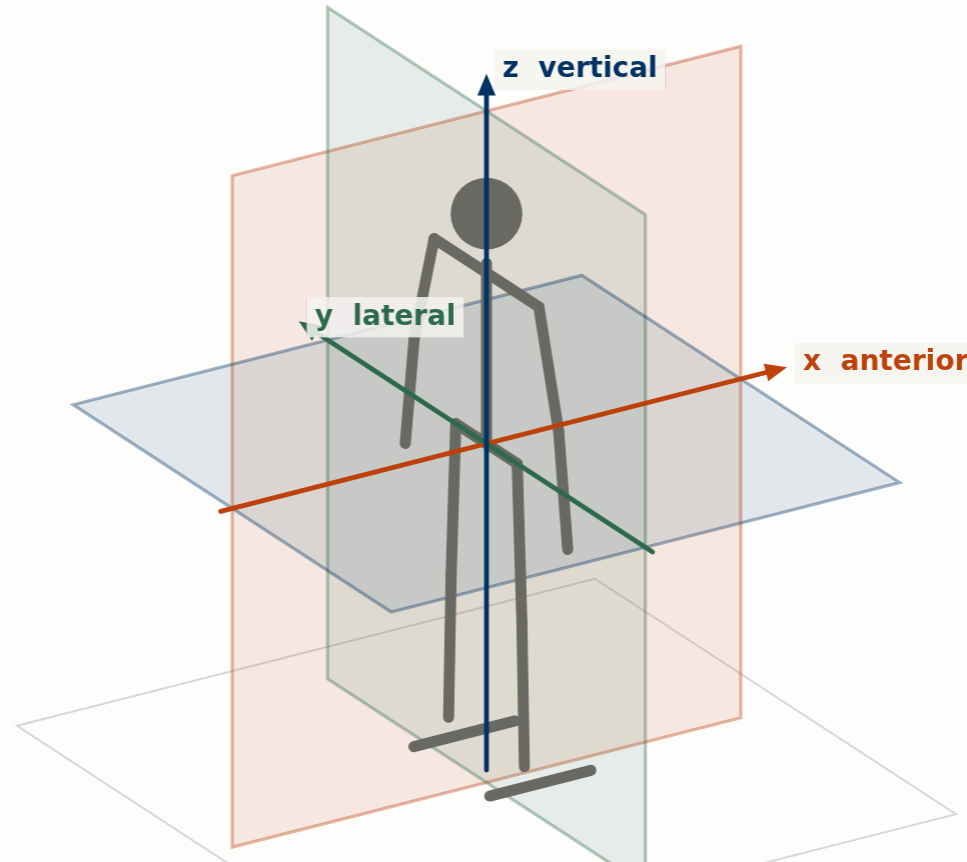
Forces in 3D

When we move we exert force in **all directions**.

Both internal and external forces are resolved into components in reference to anatomical planes and axis systems.



Three planes, three axes — spin it and pick one



Three planes, three axes.

Every force we measure gets split along these axes, and every joint movement gets named by the plane it happens in. Spin the view and pick a plane to see which axis it is built on.

When considered in a **sagittal plane**

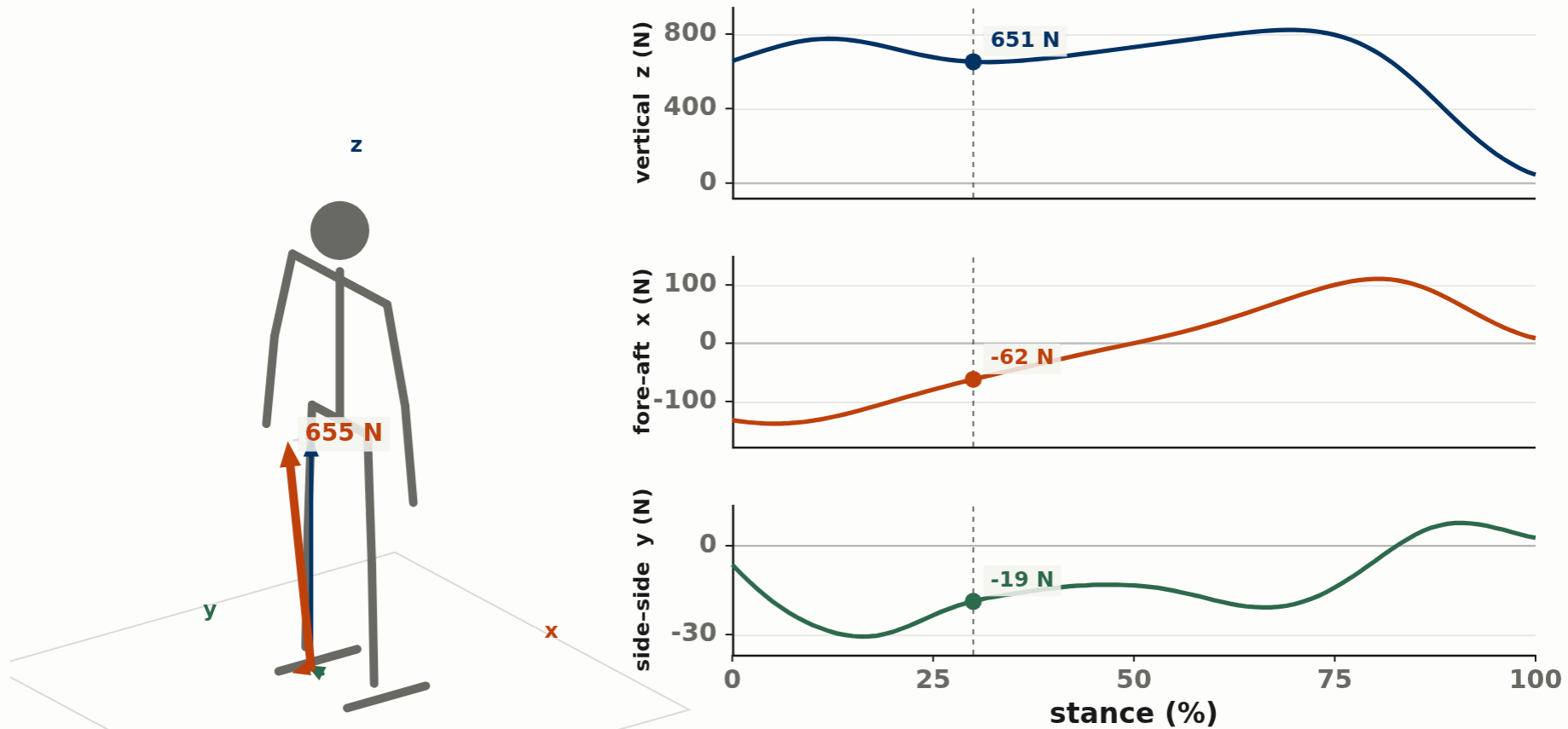
- when we start to walk we exert a force that is not just acting down (z-axis).
- part of the force is parallel to the surface (x-axis)

Also when considered in a **frontal plane**

- Part of the force is parallel to the surface acting along the y-axis



One step, measured — the same vector split into its three axes



30% of stance · vertical **651 N** (0.85 BW) · fore-aft **-62 N** · side to side **-19 N**

The fore-aft component is **negative**: the ground is pushing backwards on you. This is the braking half of the step. One walking step from a 78 kg adult, measured on a force plate (Uhlrich et al., 2023). The vertical component is roughly eight times the others, which is why it is the one everybody quotes — and why the small ones are where the interesting things hide.

Measuring external force in biomechanics

- Strain gauges
- Force Transducers
- Force plates
- Load cells
- Springs
- Piezoelectric crystals
- Dynamometer

An applied force will cause an object to deform and this deformation is measured and amplified by the electronics.



Measuring external force in biomechanics

In many clinical settings force transducers are simple **single axis** devices.

A good example is a hand-held dynamometer. This device is used to determine:

- muscle strength capabilities
- weaknesses



Measuring external force in biomechanics

Gold standard measurement of force in biomechanics is the **forceplate** and **instrumented treadmill**.

PRO'S

Accurate measure of ground reaction force

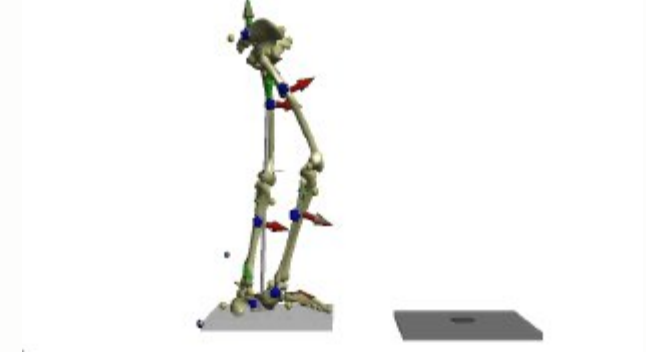
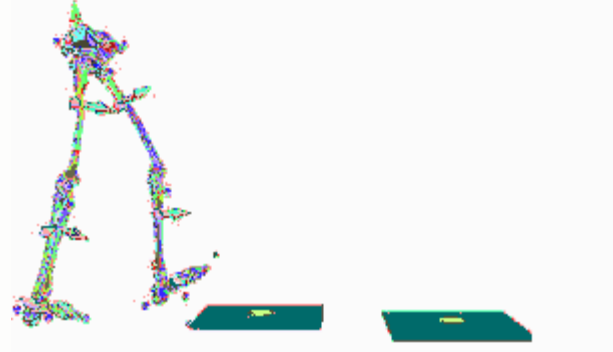
Reliable

Multidimensional

CON'S

Difficult to use in the field

Cannot measure foot pressure distribution



Measuring external force in biomechanics

Often it is not feasible to directly measure external force but an inexpensive alternate is to measure **pressure**.

Pressure

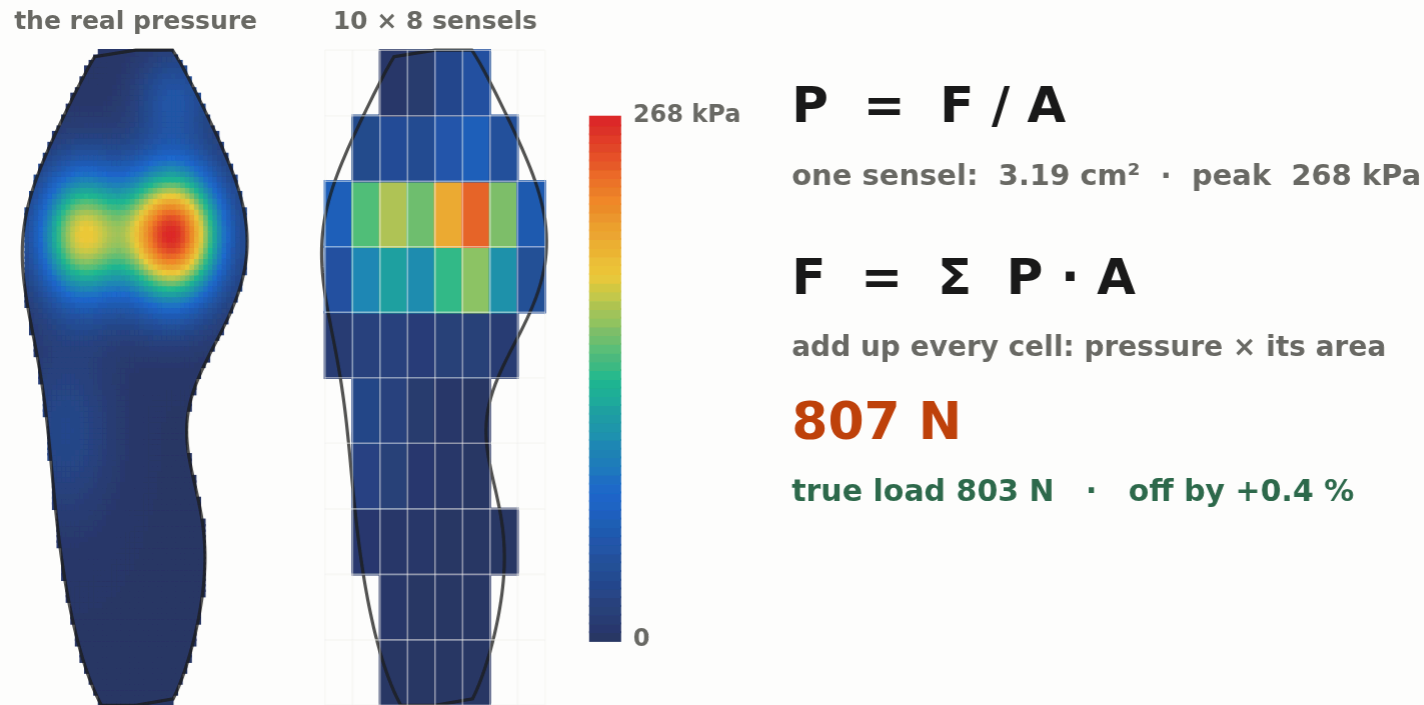
- Is the force per unit area directed perpendicular to an object surface
- $P = F/A$



Pressure Mats and Sensors

Measuring external force in biomechanics

$P = F/A$ — and how an insole gets the force back



10 × 8 sensels of 3.19 cm² · peak **268 kPa** · $\sum P \cdot A = 807 \text{ N}$ against a true 803 N

A pressure sensor never measures force. It measures pressure on a known area and you add it back up — which is why an insole is an approximation of force, and why the number of sensels matters: coarsen the grid and the peaks fall between cells.

Through stance

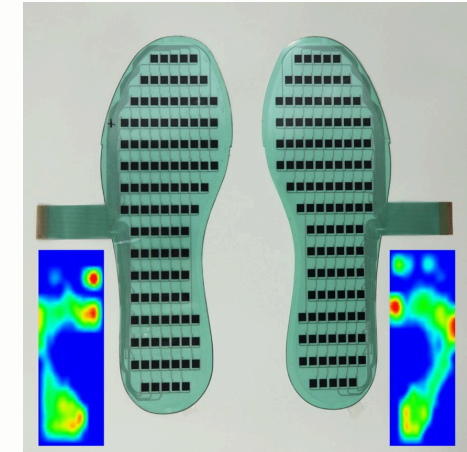
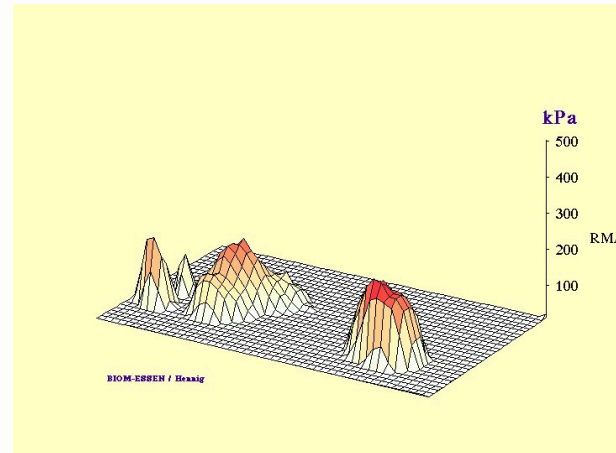
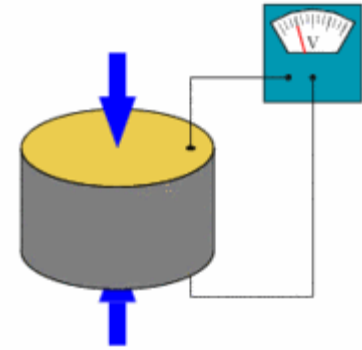
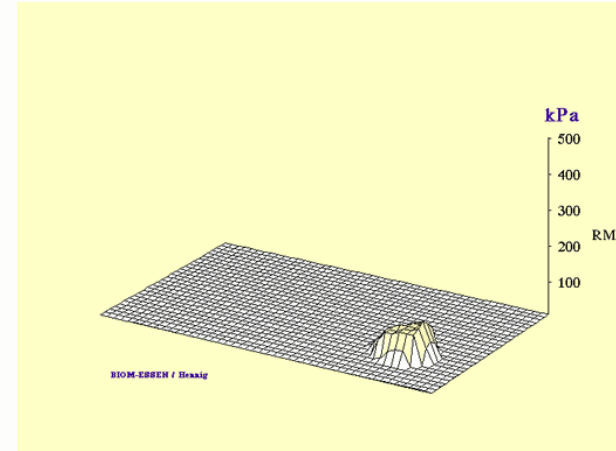


70 %

How pressure sensors work

Most common pressure sensors measure pressure using the **piezoresistive effect**.

- **Piezo** – means to squeeze or press
- The piezoresistive effect is a change in the electrical resistivity of a semiconductor or metal when mechanical strain is applied.



Measuring external force in biomechanics

A common research and field based measure of foot loading is **pressure sensing technology**.

PRO'S

Portable, thin, flexible, lightweight

Approximation of force

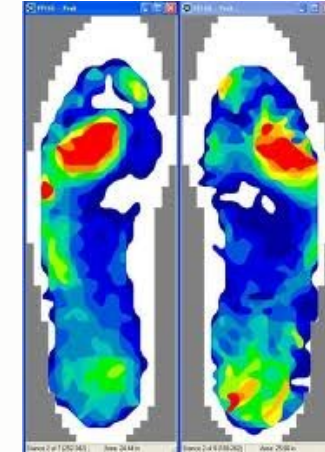
Accurate for timing of foot loading

Can measure foot pressure distribution while force plates cannot

CON'S

Unidimensional (vertical loading)

Degree of precision, accuracy and reliability requires testing – refinement



Measuring external force in biomechanics

There are many insole **plantar pressure measurement systems (PPMS)**.

These sensors help to provide the potential to measure external pressure (and calculate external force).



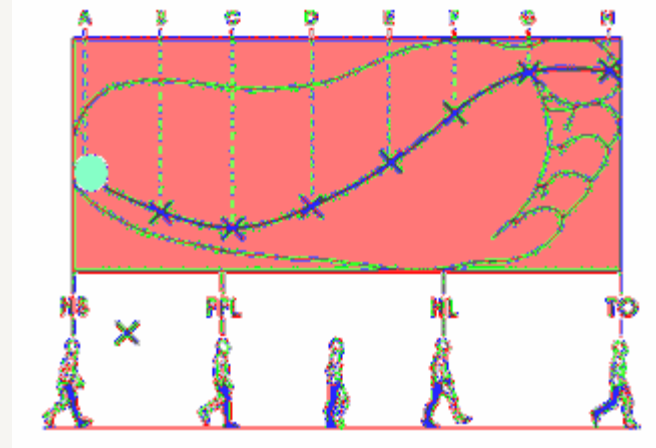
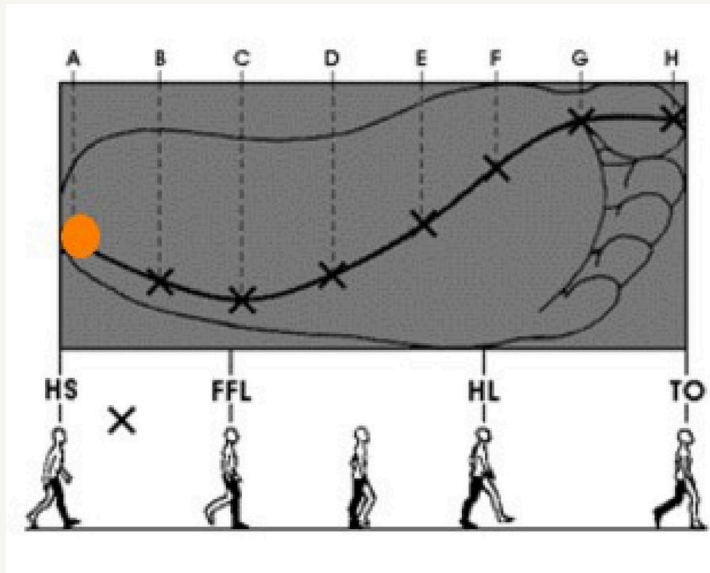
Plantar pressure sensors



Centre of Pressure

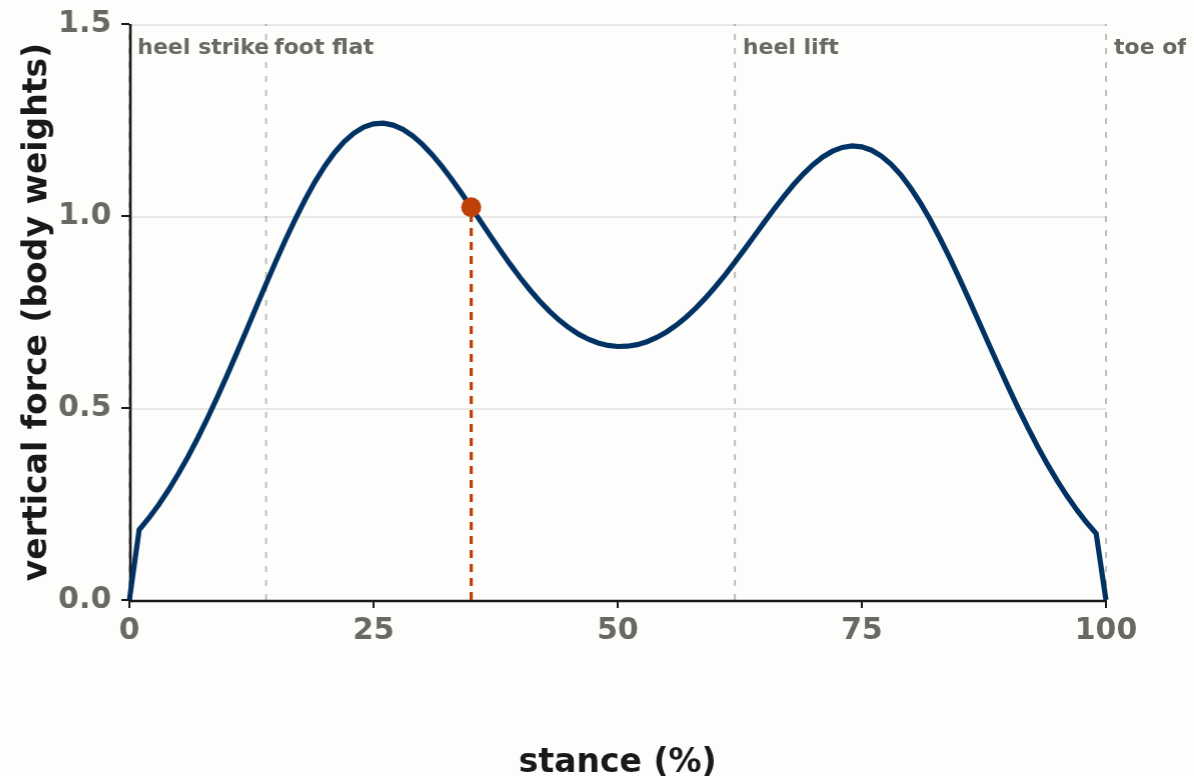
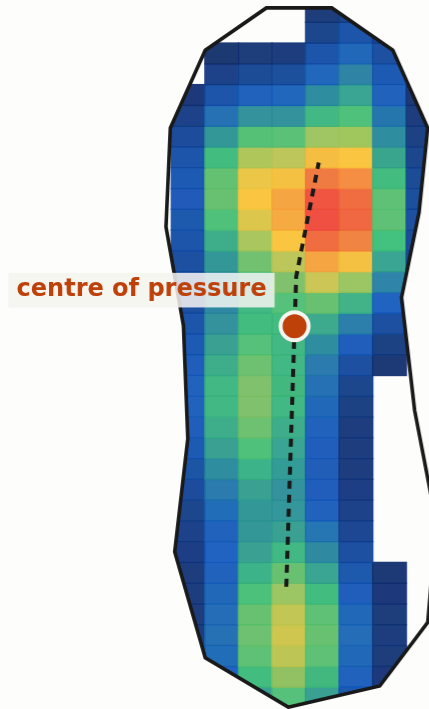
Centre of Pressure is the **instantaneous point of application** of the ground reaction force.

Measuring foot centre of pressure movement during walking can tell us more information about the GRF vector and how we use our feet while walking.



Centre of pressure

The same path, measured — where the force is applied, moment by moment



35% of stance · rolling forward through the midfoot · vertical force **1.02** body weights

The centre of pressure is the single point the ground reaction force acts through at that instant — not where the pressure is highest, but the average of the whole field. It sweeps heel to toe in about 0.6 s of walking.

Stance



35%

PPMS can tell us similar information about COP as compared to force plates.

Dyer & Morris Bamberg, Instrumented insole vs. force plate: a comparison of center of plantar pressure, IEEE.

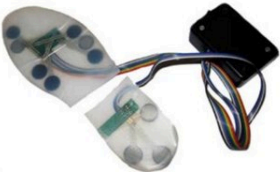
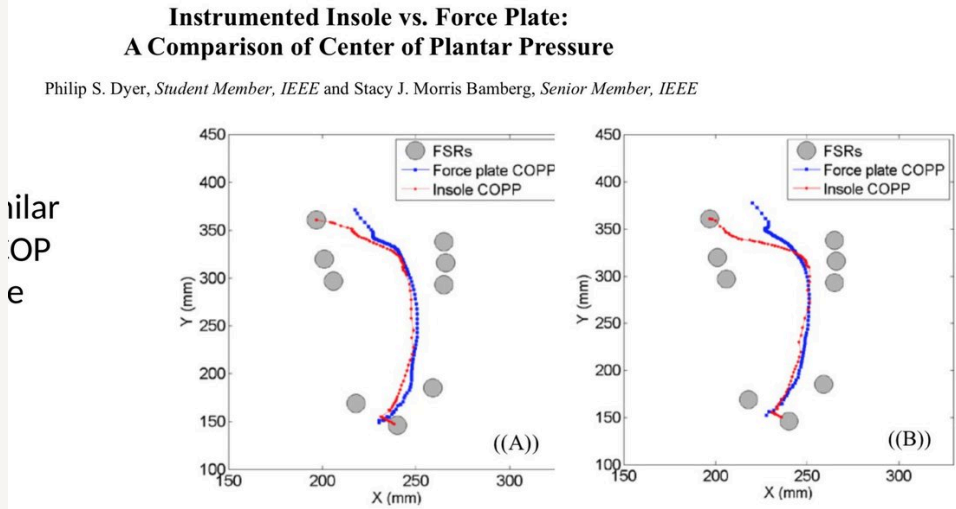


Fig. 1. Custom-built instrumented insole.

Table I Pearson Correlation Coefficients and RMS Error				
	Pearson Correlation		RMS (mm)	
	X	Y	X	Y
Min	0.68	0.97	7	13
Max	0.97	0.99	14	24
Mean	0.87 ± 0.07	0.99 ± 0.003	11 ± 1	17 ± 2

Instrumented devices

Force and pressure sensors can be added to mobility devices so that these objects become “instrumented” and we can measure the forces that individuals produce during activity.

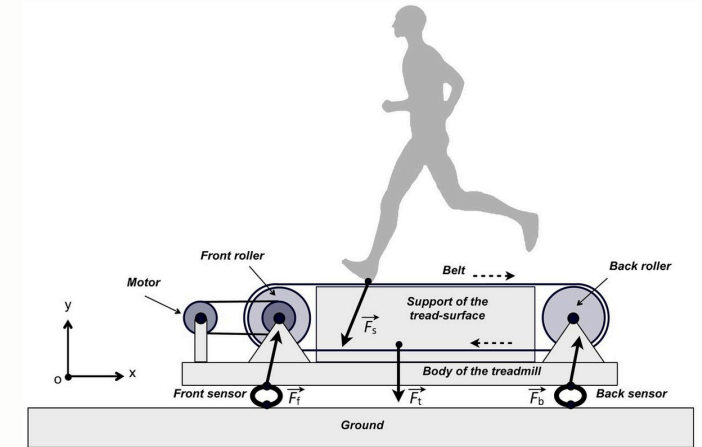


Measuring external force in biomechanics

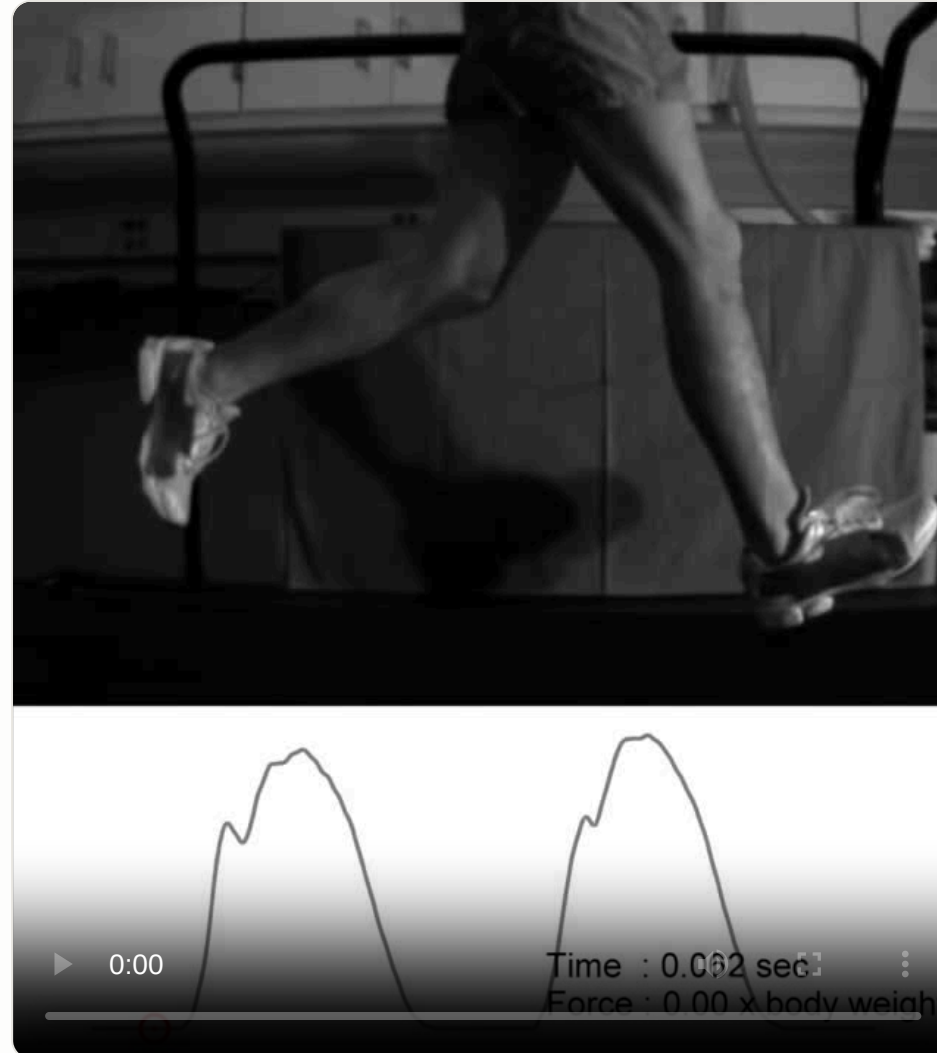
Force plate application

- Running style
 - Heel strike
 - Forefoot strike
- Footwear
 - Barefoot
 - Shoes
 - Vibram five fingers

Foot strike patterns and collision forces in habitually barefoot versus shod runners. Lieberman et al. *Nature*
nature.com/nature/journal/v463/n7280/pdf/nature08723.pdf

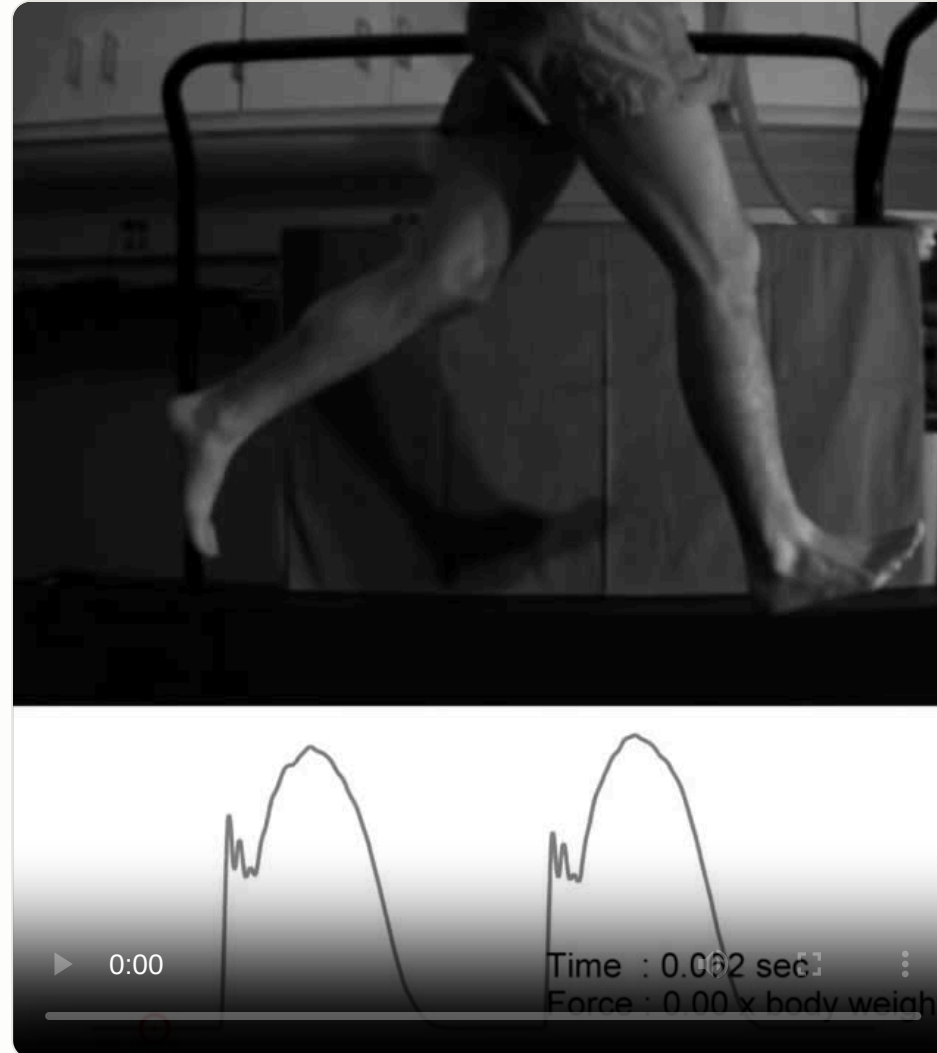


Shod heel strike



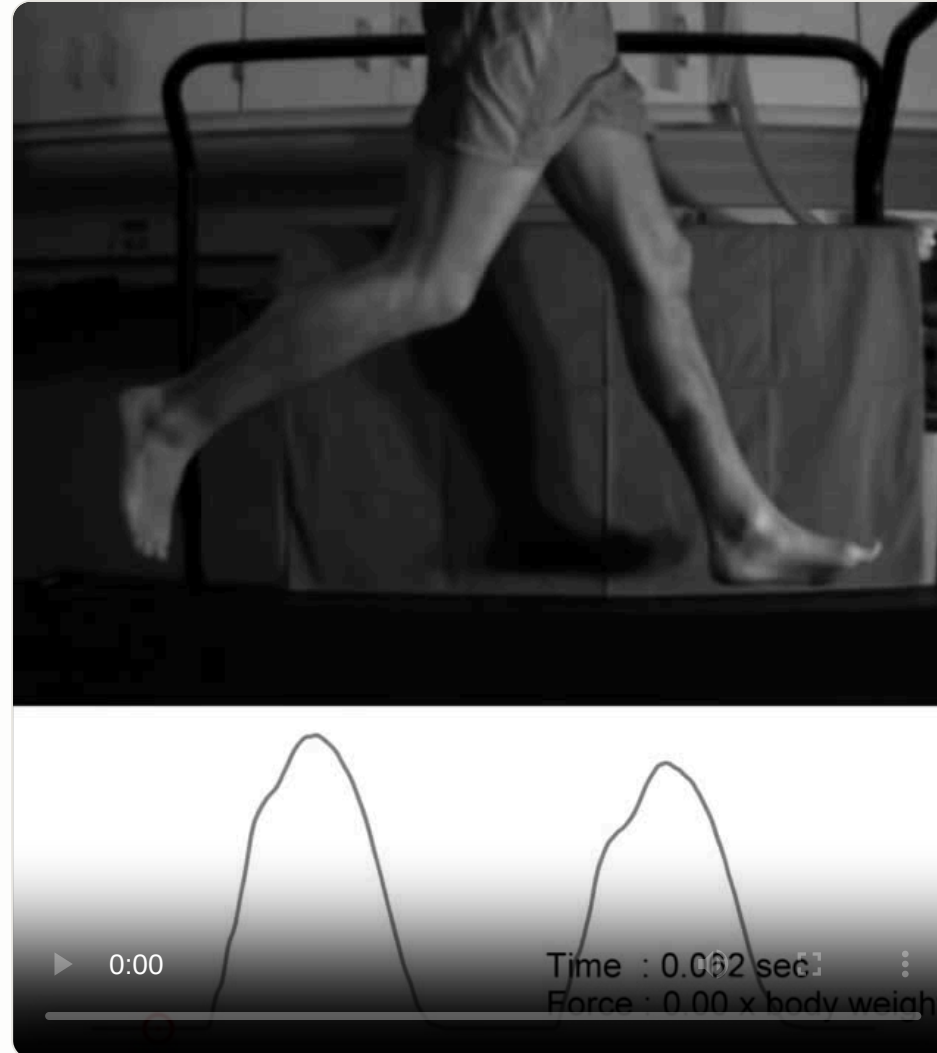
Slow motion with the vertical ground reaction force underneath.

Barefoot heel strike



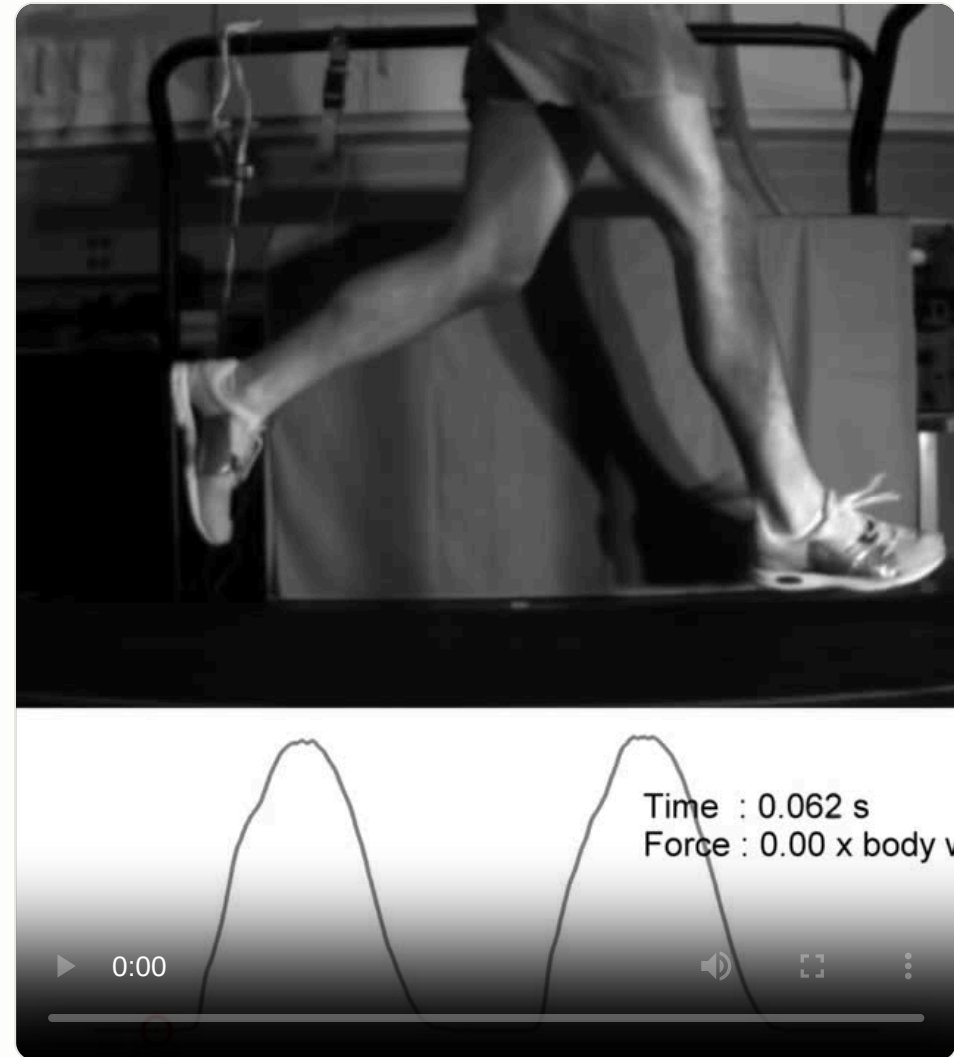
Same runner, same heel strike, no shoe.

Barefoot forefoot strike



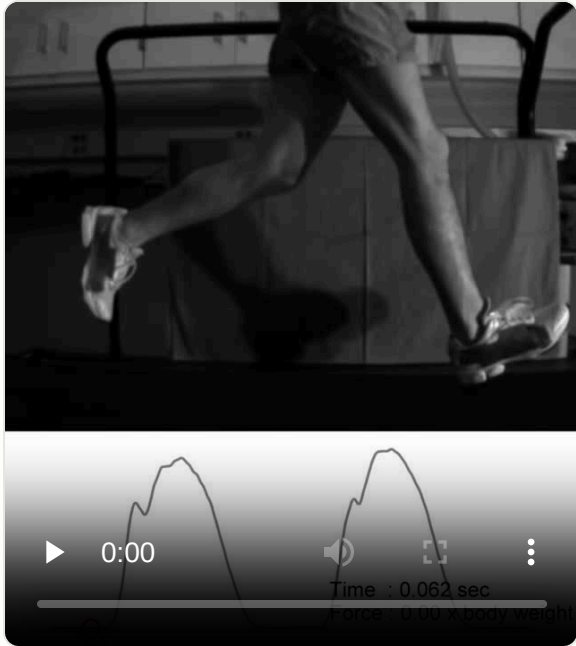
The forefoot lands first and the heel comes down afterwards.

Racing flat, forefoot strike



A forefoot strike in a thin-soled racing flat.

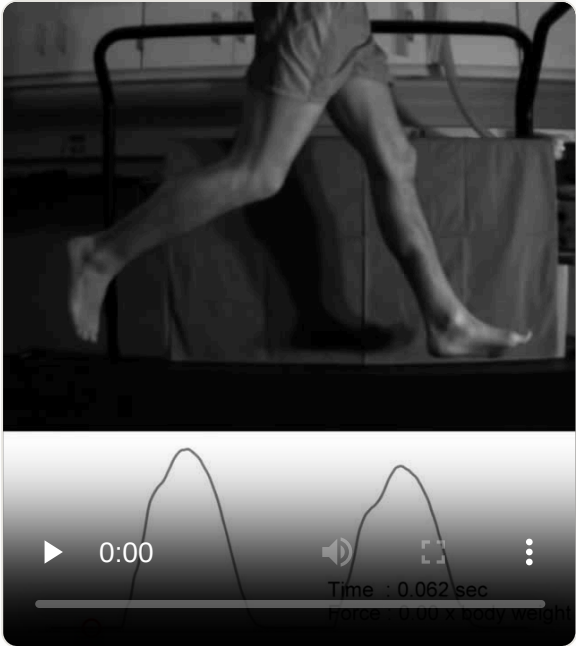
All four together



shod, heel strike



barefoot, heel strike



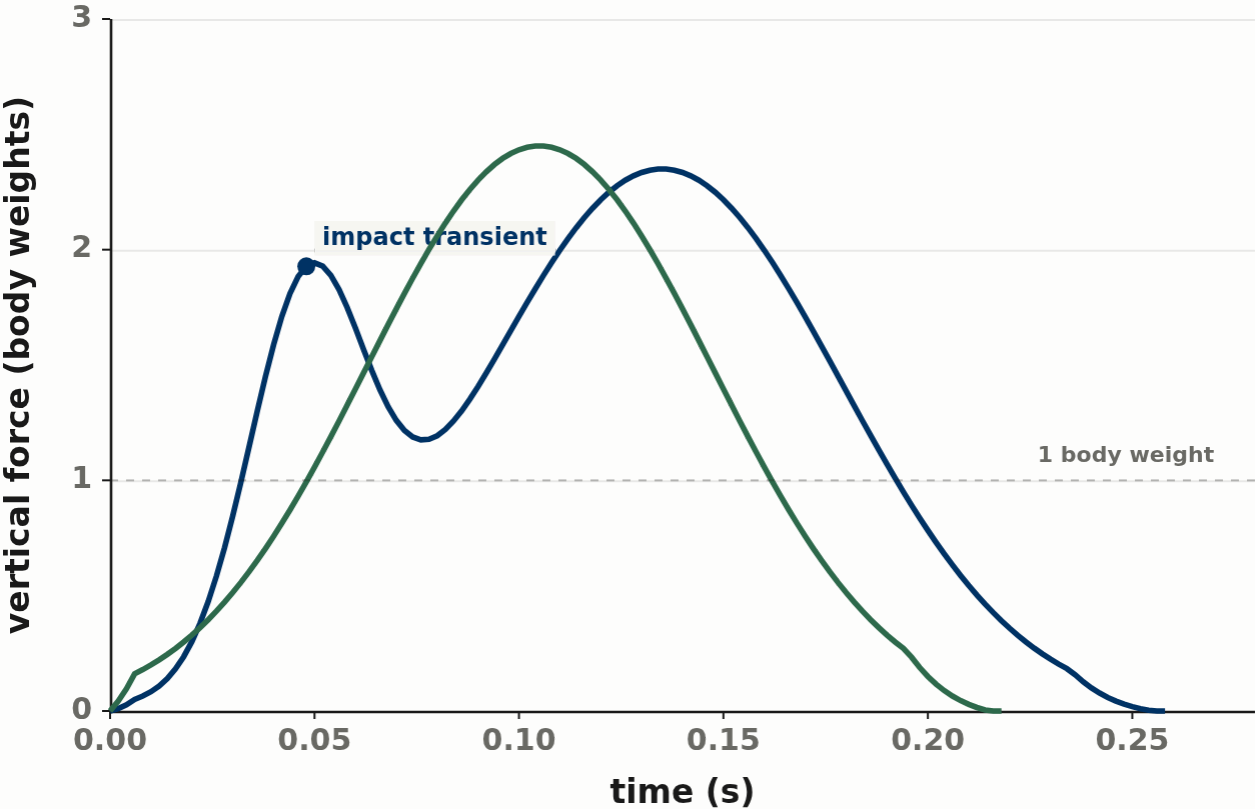
barefoot, forefoot strike



racing flat, forefoot strike

The same comparison, live

Every style lands on about the same peak force. What separates them is the spike in the first 50 milliseconds — and how fast the load arrives.



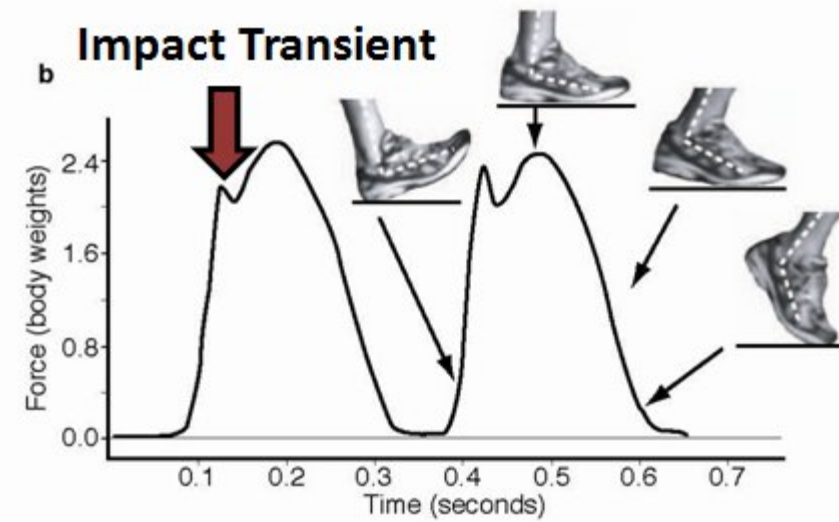
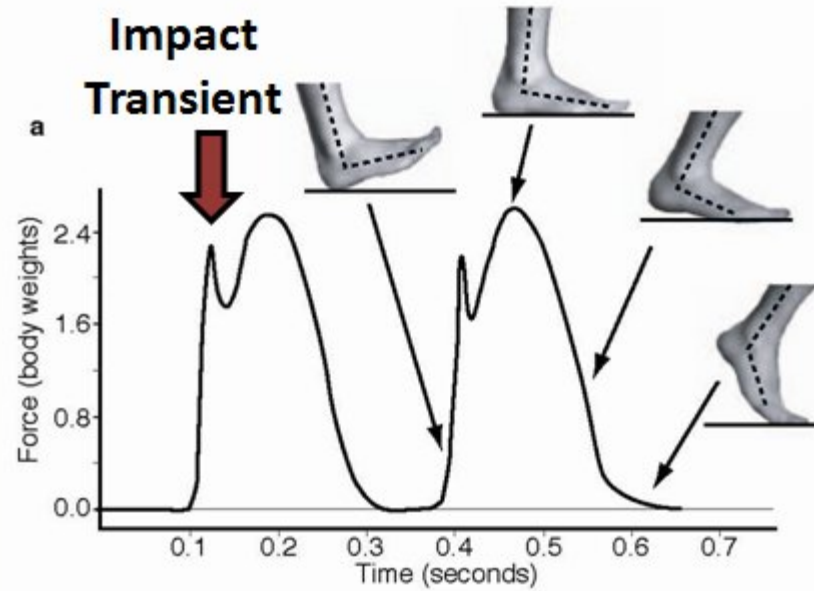
peak force and how fast it arrives

	peak	in N	rate
shod, heel strike	2.35	1614	75
barefoot, heel strike	2.71	1860	264
barefoot, forefoot	2.45	1682	35

Peak in body weights, the same force in newtons for a 70 kg runner, and the steepest rise in body weights per second.

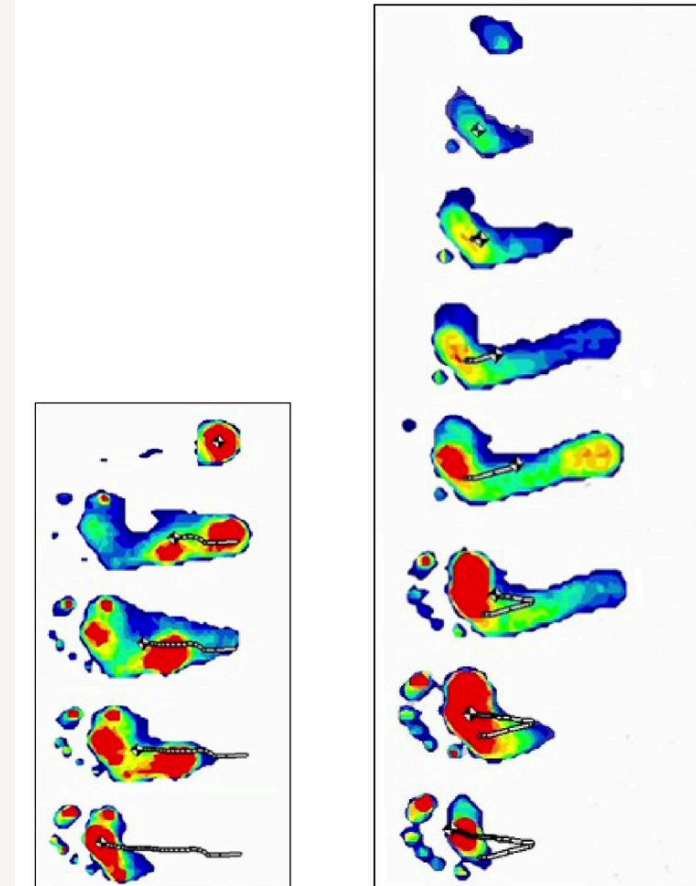
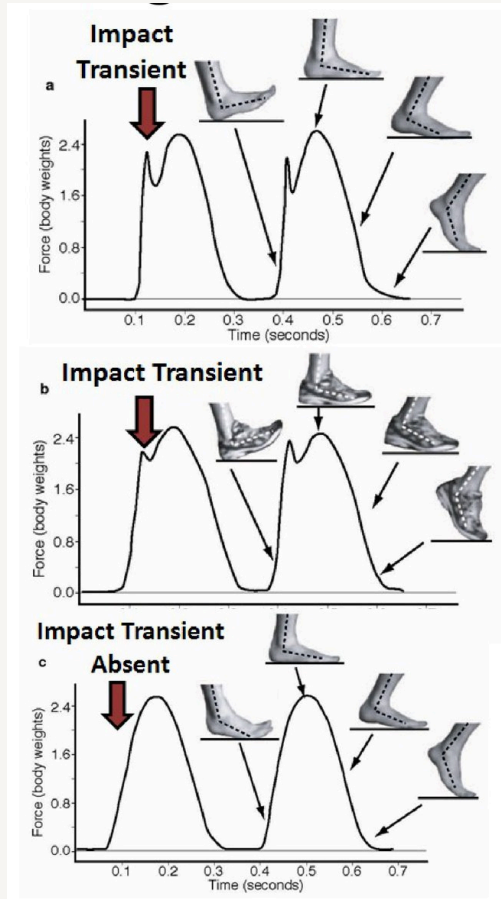
Every style ends up near the same peak. What separates them is the **spike in the first 50 ms** and how quickly the load arrives — which is the part a heel strike in a cushioned shoe softens, and a bare heel does not.

The published traces



Lieberman et al., *Nature* 463, 531–535 (2010). The arrow marks the impact transient.

Measuring external force in biomechanics



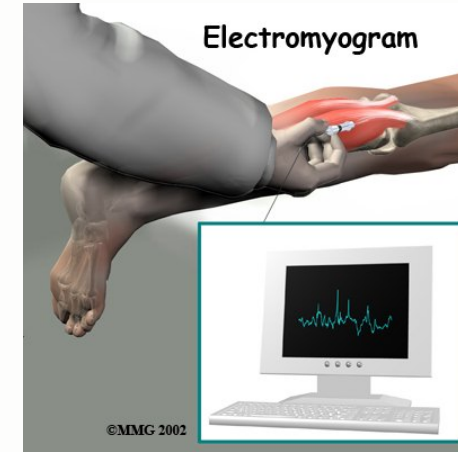
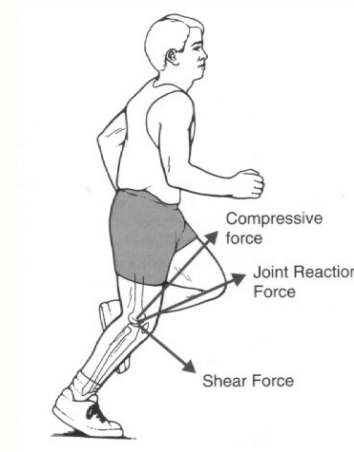
Daniel Lieberman · barefootrunning.fas.harvard.edu

Measuring internal forces in Biomechanics

While force plates and pressure sensors can be used to measure external forces, it is difficult to measure the forces at the bones, joints and tendons directly.

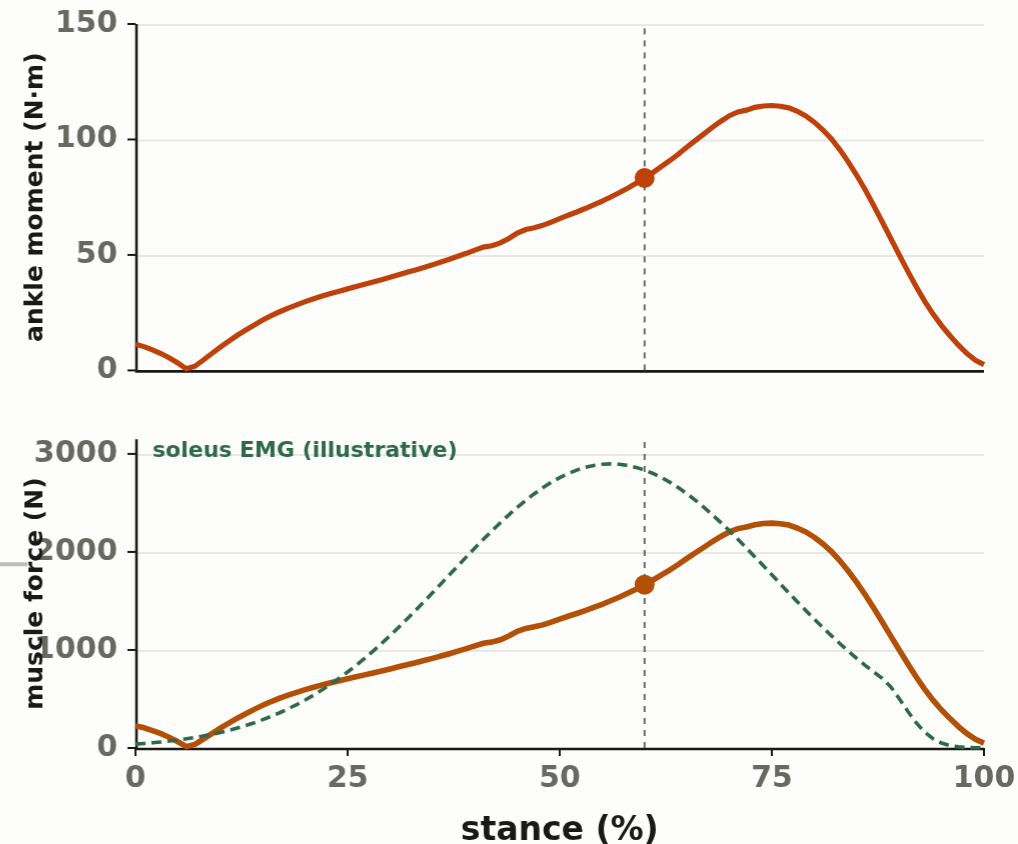
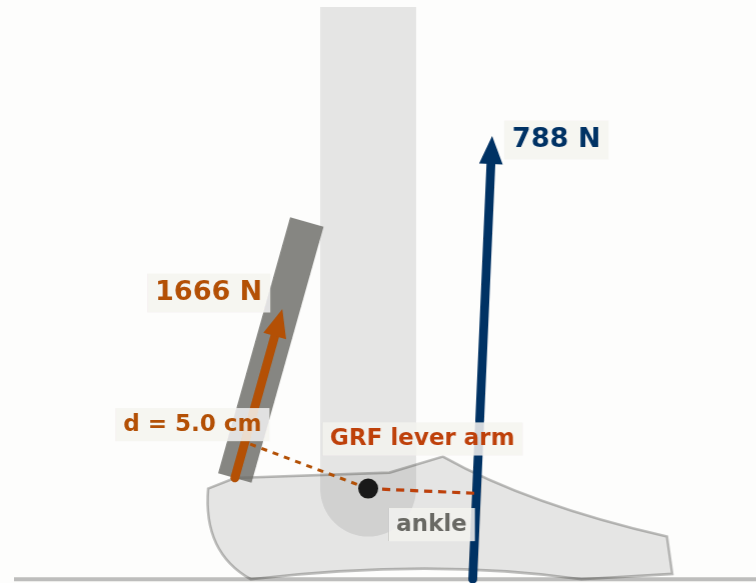
But we can use different methods to approximate the internal forces:

- inverse dynamics
- electromyography



Measuring internal forces in Biomechanics

Inverse dynamics gives the net moment — EMG says who made it



60% of stance · net ankle moment **83 N·m** ÷ moment arm **5.0 cm** = plantarflexor force **1666 N** (2.2 body weights)

Inverse dynamics works backwards from the measured ground reaction force and the segment motion to the **net** moment at the joint. Divide by the moment arm and you get a muscle force — but only if one muscle group made all of it, and only if you know the arm. Slide the arm from 3 to 7 cm and the answer halves. Moments from the measured trial (Uhlrich et al., 2023); the EMG envelopes are drawn from the textbook pattern, not from this subject.